A local Lamperti law for the occupation count of a lattice Lévy walk with infinite-mean runs

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Abstract

A lattice Lévy walk moves in runs of alternating direction whose independent lengths have regularly varying tails of index $\alpha \in (0,1)$. Let A_N be the number of right steps among the first N. Lamperti's theorem gives the limit law of A_N/N. We prove its local form. The rescaled point probabilities $N\mathbb{P}(A_N = k)$ converge to Lamperti's density at one interior point if and only if they converge uniformly on compact sets, and this happens if and only if both run laws satisfy the strong renewal theorem. By the characterization of Caravenna and Doney, the local law therefore holds with no further condition when $\alpha > 1/2$, and for $\alpha \le 1/2$ it holds exactly under their condition for each run law. We identify the limit directly from the renewal structure. For equal tails the density is U-shaped if $\cos(\pi\alpha/2) \ge \alpha$ and W-shaped otherwise, with exactly 1 or 3 critical points, which gives the threshold $\alpha = 0.5946\ldots$ in closed form. Near the ends the law is the tail of a run times a renewal mass function, and under Doney's local condition this boundary layer matches the bulk uniformly, so the global mode sits at bounded distance from an end for every α. For an arbitrary initial run we give the exact defect in the bulk, a necessary condition, a sufficient condition, and an exact criterion. An initial run that is finite almost surely can change the local law in the bulk. We also give a lower bound on the number of local extrema of the law.

Keywords: Lamperti law; generalized arcsine law; occupation time; alternating renewal process; local limit theorem; strong renewal theorem; Lévy walk; infinite mean.
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1 Introduction

A lattice Lévy walk moves on $\mathbb{Z}$ in runs of alternating direction. The run lengths are independent, the right runs have one law and the left runs another, and we assume that both tails are regularly varying with the same index $\alpha \in (0,1)$. So the runs have infinite mean. Let A_N be the number of right steps among the first N, and put $P_N(k) = \mathbb{P}(A_N =$

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k) for $0 \le k \le N$. The walk is at $2A_N - N$ after N steps, so P_N is the law of the walk on its natural lattice, and it is also the law of the occupation count of a discrete-time alternating renewal process. In the infinite-mean regime A_N/N does not concentrate. Its limit law is a generalized arcsine law, with density

$$L_{\alpha,b}(x) = \frac{\sin \pi\alpha}{\pi} \frac{b\, x^{\alpha-1}(1-x)^{\alpha-1}}{b^2 x^{2\alpha} + 2b\cos(\pi\alpha)\, x^{\alpha}(1-x)^{\alpha} + (1-x)^{2\alpha}}, \qquad 0 < x < 1, \tag{1.1}$$

where b is the limit ratio of the left tail to the right tail. This density goes back to Lamperti [13], and in the physics literature it is called the Lamperti distribution [14, 20]. This paper is about the local form of this law, at single points k.

1.1 The local Lamperti law

Write $T_{\pm}(n) = \mathbb{P}(X_{\pm} \ge n)$ for the tails of the two run lengths. A renewal law with a tail T of index $\alpha \in (0,1)$ satisfies the strong renewal theorem (SRT) if its renewal mass function u satisfies

$$u(n) \sim \frac{\sin \pi\alpha}{\pi} \frac{1}{nT(n)} \qquad (n \to \infty).$$

For $\alpha > 1/2$ this always holds under aperiodicity, by Garsia and Lamperti [9] and Erickson [7]. For $\alpha \le 1/2$ it can fail, and Caravenna and Doney [4] found the exact condition. Our first theorem says that the local Lamperti law is equivalent to the strong renewal theorem for both run laws.

Theorem A (the local Lamperti law)**.** *Assume that T_+ and T_- are regularly varying with index $-\alpha$, $0 < \alpha < 1$, that $T_-(n)/T_+(n) \to b \in (0,\infty)$, and that the run laws are aperiodic. Let the walk start with a fresh run of either sign. Then the following are equivalent.*

(i) $NP_N(\lfloor x_0 N \rfloor) \to L_{\alpha,b}(x_0)$ *for one* $x_0 \in (0,1)$.

(ii) $NP_N(k) = L_{\alpha,b}(k/N) + o(1)$ *uniformly over* k *with* k/N *in each compact subset of* $(0,1)$.

(iii) Both run laws satisfy the strong renewal theorem.

With the theorem of Caravenna and Doney this gives a complete answer. For $\alpha > 1/2$ the local law holds under regular variation and aperiodicity alone. For $\alpha \le 1/2$ it holds if and only if each run law satisfies their condition, which we recall in Lemma 3.1, and Doney's local condition $f_{\pm}(n) = O(T_{\pm}(n)/n)$ suffices. A single interior point already forces the strong renewal theorem for both laws. Without it the lower half of the local law still holds, $\liminf_N (NP_N(k) - L_{\alpha,b}(k/N)) \ge 0$ uniformly on compact sets, and the failure of the strong renewal theorem shows up as an excess at every interior point (Corollary 4.3).

The proof does not need a bivariate local renewal theorem. An exact identity writes $P_N(k)$ through the convolution powers of the two run laws, sorted by the number m of completed runs. For m of order $1/T_+(N)$ the local limit theorem of Gnedenko gives a Riemann sum for an explicit double integral. The key estimate, Lemma 3.4, bounds the probability that the $(m+1)$-th run covers a level k by $CT(k)$ uniformly in m, under regular variation alone. It decouples the two signs for small m, and the contribution of small m of each sign is then exactly the quantity that Caravenna and Doney show to be asymptotically negligible if and only if the strong renewal theorem holds. The limit comes out as a double integral over stable densities, and we identify it with $L_{\alpha,b}$ through its Stieltjes transform (Proposition 4.1). This gives a self-contained derivation of Lamperti's density in this setting.

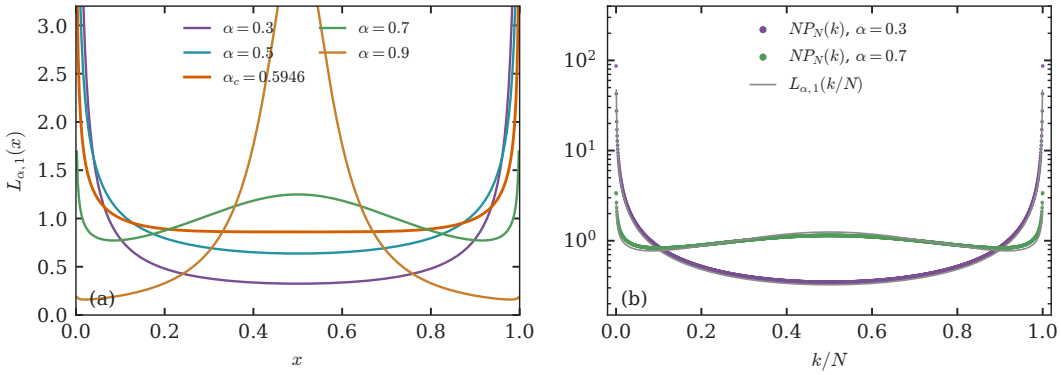


Figure 1: (a) The density $L_{\alpha,1}$ for $\alpha = 0.3$, 0.5, α_c, 0.7 and 0.9. At α_c the minimum at $1/2$ is flat to second order. (b) The exact law $NP_N(k)$ for $N = 2000$, the zeta law $f(n) \propto n^{-\alpha-1}$ for both signs and the fair start, with $\alpha = 0.3$ and $\alpha = 0.7$, on a logarithmic scale, together with $L_{\alpha,1}(k/N)$. The end values are of order $NT(N)$ and lie far above the density for $\alpha = 0.3$. Section 9 describes the computations.

1.2 The shape

For equal tails, $b = 1$, the density $L_{\alpha,1}$ is U-shaped for small α and has an interior maximum at $1/2$ for α close to 1. Margolin and Barkai [14] state that the transition happens at $\alpha = 0.5946\ldots$, and the review of Zaburdaev, Denisov and Klafter [20] says that the shape is W-like for $\alpha \gtrsim 0.6$. We give the threshold in closed form and prove the exact count.

Theorem B (the shape)**.** *$L_{\alpha,1}$ has exactly 1 critical point, a minimum at $1/2$, if $\cos(\pi\alpha/2) \geq \alpha$, that is if $\alpha \leq \alpha_c = 0.5946116440\ldots$. For $\alpha > \alpha_c$ it has exactly 3, a strict local maximum at $1/2$ and two strict minima placed symmetrically.*

The proof reduces the derivative to a function of one variable that is strictly increasing. Figure 1(a) shows the density on both sides of the threshold.

The discrete law P_N has more structure near the ends. At distance j from the plus end it is the tail $T_+(N)$ times a renewal mass function (Theorem C). Under Doney's condition this boundary layer matches the bulk uniformly up to distance $o(N)$ from the end, and $NT_+(N) \to \infty$ makes the end values much larger than the bulk values. So the global mode lies at bounded distance from an end for every $\alpha \in (0, 1)$ (Corollary 6.1). For $1 < \alpha < 2$ the same comparison has a golden ratio threshold [16], and here it has none. The ends also add local extrema of their own. Theorem E gives a lower bound on the number of sign changes of $k \mapsto P_N(k+1) - P_N(k)$, as the critical points of $L_{\alpha,b}$ plus the sign changes of the two end profiles. Conjecture 8.2 states that the bound is exact for large N. It matches every case we computed.

1.3 The initial run

Wang, Schulz, Deng and Barkai [18] studied renewal processes with fat-tailed sojourn times in continuous time and left "the effect of initial preparation on the rare fluctuations" for future work [18, Section VIII]. For the occupation count with $1 < \alpha < 2$ this is answered in [16]. Theorem D treats $0 < \alpha < 1$. Let the first run have sign σ and length d with probability $\nu_\sigma(d)$. In the finite-mean case the bulk needs no condition on the initial run on any macroscopic interior. Here it does. A first run that ends just before the k-th plus step leaves a remainder that sits in its own boundary layer, where the probabilities are of order $T(N)$ and not $1/N$. Theorem D gives the exact defect in the

bulk, the necessary condition $\nu_{\pm}(n) = o(1/(nT_{+}(n)))$, the sufficient condition $n\nu_{\pm}(n) \to 0$ under Doney's condition, and an exact criterion. It also shows by example that an initial run which is finite almost surely can change the local law at interior points, and that the necessary condition is not sufficient. At the ends the criterion becomes the window condition $\sum_{m<N/2} \nu_{+}(N-m)T_{+}(m) = o(T_{+}(N))$. Under Doney's condition every fixed initial run with $n\nu_{\pm}(n) \to 0$ leaves the local law and the end layers unchanged. An initial run of length of order N should change the local law itself, by an explicit mixture, and we state this as Conjecture 7.1.

1.4 Related work

Occupation times. Lamperti [13] proved an occupation time theorem for a class of stochastic processes, and the generalized arcsine density $L_{\alpha,b}$ is attached to his name [14, 20]. Mitov and Yanev [15] study limit theorems for the occupation-time process of alternating renewal processes when the mean cycle is infinite, extending results of Dynkin, Lamperti and Erickson for the spent lifetime. Starreveld, Bekker and Mandjes [17] give the transform of the occupation time of an alternating renewal process at an exponential time, a central limit theorem under finite second moments, large deviations, and applications to Lévy processes. Godrèche and Luck [10] give the transform of the occupation time and its limit laws. All of these are statements in distribution. We found no local limit theorem for the occupation count of an alternating renewal process with infinite means.

Strong renewal theorems. The strong renewal theorem for $\alpha > 1/2$ is due to Garsia and Lamperti [9], Williamson [19] and Erickson [7]. Doney [6] proved it for all $\alpha \in (0,1)$ under the local condition $f(n) = O(T(n)/n)$, together with a local large deviation bound. Caravenna and Doney [4] gave the necessary and sufficient condition for $\alpha \leq 1/2$, and their local large deviation bound for sums of a fixed number of variables is the main input of our key estimate. Berger [2] extends Doney's local bound, and his version is the one we quote. Berger [1] also proved multivariate strong renewal theorems. Theorem A shows that for the occupation count the bivariate problem reduces exactly to the two univariate ones.

Physics. In the physics literature the walk is the Lévy walk in the ballistic regime [20], and the occupation count is the time average of a blinking signal with power-law sojourn times [14]. Both sources use $L_{\alpha,1}$ as the limit density of the time average. Wang, Schulz, Deng and Barkai [18, Section VII.A] found an atom with a boundary layer for the occupation time when $0 < \alpha < 1$, for one sojourn law and a start of fair random sign, through a double Laplace transform. Theorem C is a lattice version of this layer at a fixed time, for two laws, with the uniform matching to the bulk.

The finite-mean case. For $1 < \alpha < 2$ the occupation law has a stable bulk, big-jump zones and boundary layers, and its initial-run dependence and modes are described in [16]. We use the exact identities of that paper and recall them with proofs in Section 2.5.

Organization. Section 2 fixes notation and proves the exact identities. Section 3 collects the cited theorems and proves the elementary bounds, including the key estimate. Section 4 proves Theorem A, Section 5 proves Theorem B, Section 6 treats the ends and the global mode, Section 7 treats the initial run, and Section 8 counts sign changes. Section 9 describes the computations. No proof uses a computed number.

2 Setting, hypotheses and exact identities

2.1 Runs and the occupation count

The walk takes N steps and each step has a sign, $+$ or $-$. The signs come in maximal runs of equal sign, and consecutive runs have opposite signs. All runs after the first are independent. A run of sign $\sigma \in \{+,-\}$ has a length distributed as X_σ, a random variable on $\{1,2,\dots\}$ with mass function $f_\sigma(n) = \mathbb{P}(X_\sigma = n)$ and tail

$$T_\sigma(x) = \mathbb{P}(X_\sigma \geq x), \qquad x \in \mathbb{R}.$$

We write $\bar\sigma$ for the sign opposite to σ. The first run is the initial run. It has a sign Σ and a length D, the pair (Σ, D) is independent of all later runs, and $D < \infty$ almost surely. We put

$$\nu_\sigma(d) = \mathbb{P}(\Sigma = \sigma,\, D = d), \qquad Q_\sigma(N) = \mathbb{P}(\Sigma = \sigma,\, D \geq N), \qquad \pi_\sigma = \mathbb{P}(\Sigma = \sigma).$$

Except in Conjecture 7.1, the law of the initial run does not depend on N. A fresh start of sign σ_0 has $\nu_{\sigma_0} = f_{\sigma_0}$ and $\nu_{\bar\sigma_0} = 0$. The fair start is a fresh start of random sign, each sign with probability $\frac{1}{2}$.

Let A_N be the number of plus steps among steps $1,\dots,N$, and $P_N(k) = \mathbb{P}(A_N = k)$ for $0 \leq k \leq N$. For a fresh start of sign σ we write $F_N^\sigma(k)$ for $P_N(k)$, with $F_N^\sigma(k) = 0$ for $k \notin \{0,\dots,N\}$. We also use

$$\Pi_m^-(j) = F_m^-(m-j),$$

the probability that a fresh minus walk has exactly j minus steps among its first m steps.

2.2 Hypotheses

The hypotheses carry fixed names, and every statement says which it uses.

(RV$_\alpha$) T_+ is regularly varying with index $-\alpha$, where $0 < \alpha < 1$, and $T_-(n)/T_+(n) \to b$ for some $b \in (0,\infty)$.

(SPAN) For each sign, the support of X_σ is not contained in any arithmetic progression $c + h\mathbb{Z}$ with $h \geq 2$.

(SRT$_\sigma$) $u_\sigma(n) \sim \dfrac{\sin \pi\alpha}{\pi} \dfrac{1}{nT_\sigma(n)}$ as $n \to \infty$, with u_σ the renewal mass function of Section 2.4.

(LB$_\sigma$) There is C with $f_\sigma(n) \leq C\, T_\sigma(n)/n$ for all $n \geq 1$.

Under (RV$_\alpha$) both tails are regularly varying with index $-\alpha$. Unequal indices, or $b \in \{0,\infty\}$, give a degenerate limit of A_N/N and are not treated. (SPAN) is the aperiodicity needed for local limit theorems. (SRT$_\sigma$) is the strong renewal theorem for the runs of sign σ. (LB$_\sigma$) is Doney's local condition. It implies (SRT$_\sigma$) by Lemma 3.1(iii) below. A hypothesis written with the subscript $\pm$ is assumed for both signs.

Constants. Unindexed C and c are positive constants that depend only on the laws of the runs and may change from one use to the next. A constant written C_δ also depends on δ.

2.3 Scales, stable densities and the Lamperti density

Under (RV$_\alpha$) choose an increasing function $a : (0,\infty) \to (0,\infty)$ with $t\,T_+(a(t)) \to 1$ as $t \to \infty$, and put $a_m = a(m)$. So a is regularly varying with index $1/\alpha$. For $t > 0$ let

$$m_N(t) = \lfloor t/T_+(N) \rfloor.$$

Then $T_+(a_{m_N(t)}) \sim T_+(N)/t$, and $a_{m_N(t)}/N \to t^{1/\alpha}$ uniformly for t in compact subsets of $(0,\infty)$ (Lemma A.1).

Put $\Gamma_\alpha = \Gamma(1-\alpha)$, and let g be the density of the positive stable variable Z with $\mathbb{E}e^{-\lambda Z} = e^{-\Gamma_\alpha \lambda^\alpha}$. Its Lévy measure is $\alpha r^{-\alpha-1}dr$ on $(0,\infty)$, with tail $r^{-\alpha}$. For $t>0$ let

$$p_t(z) = t^{-1/\alpha} g\big(z t^{-1/\alpha}\big),$$

the density at time t of the stable subordinator with Laplace exponent $t\Gamma_\alpha\lambda^\alpha$. The covering densities are

$$Q_t(x) = \int_0^x p_t(z)(x-z)^{-\alpha}\,dz, \qquad Q_t^b(y) = b\int_0^y p_{bt}(z)(y-z)^{-\alpha}\,dz. \tag{2.1}$$

The density g is continuous and bounded, and $g(w) \sim \alpha w^{-1-\alpha}$ as $w\to\infty$ [8, Section XVII.6]. Hence

$$p_t(z) \le C\min\big\{t^{-1/\alpha},\, t\,z^{-1-\alpha}\big\} \qquad (t,z>0). \tag{2.2}$$

For $0<x<1$ let

$$\Lambda(x) = \int_0^\infty \big[p_{bt}(1-x)\,Q_t(x) + p_t(x)\,Q_t^b(1-x)\big]\,dt. \tag{2.3}$$

The Lamperti density $L_{\alpha,b}$ is defined in (1.1). With $t = \log((1-x)/x)$ and $\beta = \log b$ it reads

$$L_{\alpha,b}(x) = \frac{\sin\pi\alpha}{\pi}\,\frac{1+\cosh t}{\cosh(\alpha t-\beta)+\cos\pi\alpha}. \tag{2.4}$$

To check (2.4), write $w = (1-x)/x = e^t$. Then $x^{\alpha-1}(1-x)^{\alpha-1} = x^{2\alpha-2}w^{\alpha-1}$, the denominator of (1.1) is $x^{2\alpha}bw^\alpha\cdot 2[\cosh(\alpha t-\beta)+\cos\pi\alpha]$, and $x^{-2}w^{-1} = (1+w)^2/w = 2+2\cosh t$. We also use, for $0\le y<1$,

$$\Phi_b(y) = \frac{b^2(1-y)^{\alpha-1}}{b^2(1-y)^{2\alpha} + 2b\cos(\pi\alpha)\,(1-y)^\alpha y^\alpha + y^{2\alpha}}, \tag{2.5}$$

so $\Phi_b(0)=1$, $\Phi_b(y) = 1+O(y^\alpha)$ as $y\to 0$, $\Phi_b>0$, and $L_{\alpha,b}(1-y) = \frac{\sin\pi\alpha}{\pi b}\,y^{\alpha-1}\Phi_b(y)$.

2.4 Renewal kernels

For each sign let $X_1^\sigma, X_2^\sigma, \dots$ be independent copies of X_σ, $S_m^\sigma = X_1^\sigma+\cdots+X_m^\sigma$ with $S_0^\sigma = 0$, and $u_m^\sigma = f_\sigma^{*m}$ the law of S_m^σ, so $u_0^\sigma = \delta_0$. The renewal mass function is

$$u_\sigma(j) = \sum_{m\ge 0} u_m^\sigma(j), \qquad j\ge 0,$$

so $u_\sigma(0)=1$. For $m\ge 1$ and $k\ge 1$ let

$$q_m^\sigma(k) = \mathbb{P}\big(S_{m-1}^\sigma < k \le S_m^\sigma\big) = \sum_{r<k} u_{m-1}^\sigma(r)\,T_\sigma(k-r), \tag{2.6}$$

the probability that the m-th run of sign σ covers level k in the time of its sign. The second form holds because X_m^σ is independent of S_{m-1}^σ. For each $k\ge 1$ exactly one m has $S_{m-1}^\sigma < k \le S_m^\sigma$, so $\sum_{m\ge1} q_m^\sigma(k) = 1$. We write $\omega_\sigma(i) = \min\{1, 1/(iT_\sigma(i))\}$ for $i\ge 1$ and $\omega_\sigma(0)=1$. Under (SRT$_\sigma$) we have $u_\sigma \le C\omega_\sigma$.

2.5 Exact identities

The following identities hold for every N and need no hypothesis. They are proved in [16, Section 2], and we include the short proofs. The computations of Section 9 check them against a sum over all sign strings.

Proposition 2.1 (the fresh-plus identity)**.** *For $k, j \geq 1$ with $k + j = N$,*

$$F_N^+(k) = \sum_{m\geq 1} u_m^-(j)\, q_{m+1}^+(k) + \sum_{m\geq 1} u_m^+(k)\, q_m^-(j), \tag{2.7}$$

and $F_N^+(N) = T_+(N)$, $F_N^+(0) = 0$. Exchanging the signs, for $k, j \geq 1$, $F_N^-(k) = \sum_{m\geq 1}[u_m^+(k)q_{m+1}^-(j) + u_m^-(j)q_m^+(k)]$, with $F_N^-(0) = T_-(N)$ and $F_N^-(N) = 0$.

Proof. For a fresh plus start the i-th plus run fills the steps in $(S_{i-1}^+ + S_{i-1}^-,\, S_i^+ + S_{i-1}^-]$ and the i-th minus run fills those in $(S_i^+ + S_{i-1}^-,\, S_i^+ + S_i^-]$. These intervals partition $\{1, 2, \dots\}$. If step N lies in the i-th plus run, the first N steps contain exactly S_{i-1}^- minus steps. So $A_N = k$ means $S_{i-1}^- = j$, and then step N lies in that run exactly when $S_{i-1}^+ < k \leq S_i^+$. The two events concern independent sequences, so this case has probability $u_{i-1}^-(j)q_i^+(k)$. If step N lies in the i-th minus run, then $A_N = k$ means $S_i^+ = k$, and step N lies in that run exactly when $S_{i-1}^- < j \leq S_i^-$, with probability $u_i^+(k)q_i^-(j)$. The cases are disjoint. For $j \geq 1$ the plus case with $i = 1$ vanishes, and $m = i - 1$ in the plus case and $m = i$ in the minus case give (2.7). For $k = N$ only the plus case with $i = 1$ survives, with probability $T_+(N)$, and $k = 0$ is impossible. The fresh minus start is the same with the signs exchanged. □

Proposition 2.2 (the first-run decomposition)**.** *For every initial run and every $0 \leq k \leq N$,*

$$P_N(k) = Q_+(N)\mathbf{1}_{\{k=N\}} + Q_-(N)\mathbf{1}_{\{k=0\}} + \sum_{d<N} \nu_+(d)\, F_{N-d}^-(k-d) + \sum_{d<N} \nu_-(d)\, F_{N-d}^+(k).$$

In particular $P_N(N) = Q_+(N)$ and $P_N(0) = Q_-(N)$.

Proof. We condition on the initial run. If $\Sigma = +$ and $D \geq N$, then $A_N = N$. If $\Sigma = +$ and $D = d < N$, steps $d+1, \dots, N$ are the first $N - d$ steps of a fresh minus walk independent of (Σ, D), so $A_N = d + A'$ with A' of law F_{N-d}^-. The case $\Sigma = -$ is the same. For $d < N$ Proposition 2.1 gives $F_{N-d}^-(N-d) = 0$ and $F_{N-d}^+(0) = 0$, so only the atoms survive at $k = N$ and $k = 0$. □

Proposition 2.3 (the final-run identity)**.** *For a fresh plus start, $N \geq 1$ and $0 \leq j < N$,*

$$\begin{aligned} P_N(N-j) = &\sum_{r=0}^{j} u_r^-(j)\, \mathbb{E}\big[T_+(N-j-S_r^+);\, S_r^+ \leq N-j-1\big] \\ &+ \sum_{m=1}^{j} T_-(m) \sum_{r=0}^{j-m} u_r^-(j-m)\, u_{r+1}^+(N-j). \end{aligned} \tag{2.8}$$

Proof. We split according to the run that contains step N, the final run. The $(r+1)$-th plus run starts at step $S_r^+ + S_r^- + 1$. It is the final run and $A_N = N - j$ exactly when $S_r^- = j$ and $X_{r+1}^+ \geq N - j - S_r^+ \geq 1$. This gives the term r of the first sum. The $(r+1)$-th minus run starts at step $S_{r+1}^+ + S_r^- + 1$. If it is the final run and starts at step $N - m + 1$, then $A_N = N - j$ means $S_{r+1}^+ = N - j$ and $S_r^- = j - m$, which needs $1 \leq m \leq j$, and $X_{r+1}^- \geq m$. This gives the second sum. Exactly one run is final, and $u_r^-(i) = 0$ for $r > i$. □

For the computations we use a third form of the law. For a fresh plus start let $A(k,j)$ and $B(k,j)$ be the probabilities that a plus run, respectively a minus run, starts right after exactly k plus steps and j minus steps. A plus run is preceded by m runs of each sign and a minus run by m plus runs and $m-1$ minus runs, so

$$A(k,j)=\sum_{m\ge 0} u_m^+(k)u_m^-(j), \qquad B(k,j)=\sum_{m\ge 1} u_m^+(k)u_{m-1}^-(j).$$

Splitting on the start of the final run gives, with $j=N-k$,

$$F_N^+(k)=\sum_{k'<k} A(k',j)\,T_+(k-k')+\sum_{j'<j} B(k,j')\,T_-(j-j'). \tag{2.9}$$

Every term is nonnegative.

3 Tools

3.1 Regular variation

We use the following facts about a function T that is regularly varying with index $-\alpha\in(-1,0)$, all from [3, Sections 1.2 to 1.5].

(R1) (Uniform convergence.) $T(\lambda x)/T(x)\to\lambda^{-\alpha}$ uniformly for λ in compact subsets of $(0,\infty)$.

(R2) (Potter.) For every $c\ge 1$ there is C_c with $T(x/c)\le C_c T(x)$ for $x\ge c$.

(R3) (Karamata.) $\sum_{n\le w}T(n)\sim wT(w)/(1-\alpha)$ and $\sum_{1\le i\le w}1/(iT(i))\sim 1/(\alpha T(w))$ as $w\to\infty$.

(R4) (Dyadic sums.) $\sum_{l\ge 1,\,2^l\le x}2^l T(2^{l-1})\le CxT(x)$ and $\sum_{l\ge 1,\,2^l>x}T(2^{l-1})\le CT(x)$ for $x\ge 1$.

(R4) follows from (R2) and Potter's theorem [3, Theorem 1.5.6], since $wT(w)$ is regularly varying with the positive index $1-\alpha$ and T with the negative index $-\alpha$, so both sums are dominated by geometric series (Lemma A.1). Under (RV$_\alpha$) the two tails are comparable, $cT_+\le T_-\le C\,T_+$, and $mT_+(a_m)\le C$ for all $m\ge 1$. We also use $a_{\lceil m/3\rceil}\ge c\,a_m$, which follows from (R2) for the function a.

3.2 Cited theorems

Under (RV$_\alpha$) and (SPAN), S_m^+/a_m converges in law to Z, and S_m^-/a_m converges to $b^{1/\alpha}Z$, whose density is $g_b(z)=b^{-1/\alpha}g(zb^{-1/\alpha})$. Note that $t^{-1/\alpha}g_b(yt^{-1/\alpha})=p_{bt}(y)$.

Lemma 3.1 (classical inputs)**.** *Assume (RV$_\alpha$) and (SPAN).*

(i) (Gnedenko's local limit theorem [11, Theorem 4.2.1].) $\sup_x|a_m u_m^+(x)-g(x/a_m)|\to 0$ *and* $\sup_y|a_m u_m^-(y)-g_b(y/a_m)|\to 0$ *as* $m\to\infty$*. Hence* $\sup_x u_m^\sigma(x)\le C/a_m$ *for all* $m\ge 1$ *and both signs.*

(ii) (Local large deviations [4, Theorem 1.1, Eq. (1.6)].) For both signs there is C *with* $u_m^\sigma(x)\le C\,\dfrac{m}{a_m}\,T_\sigma(x)$ *for all* $m\ge 1$ *and* $x\ge 1$*.*

(iii) (The strong renewal theorem.) If $\alpha>1/2$*, (SRT$_\sigma$) holds for both signs [9, 7], see [4, Theorem 1.4]. If* $\alpha\le 1/2$*, (SRT$_\sigma$) holds if and only if*

$$\lim_{\delta\to 0}\limsup_{x\to\infty}\,xT_\sigma(x)\sum_{1\le z\le\delta x}\frac{f_\sigma(x-z)}{zT_\sigma(z)^2}=0$$

[4, Theorem 1.4]. In every case (LB$_\sigma$) implies (SRT$_\sigma$) [6], see [4, Eq. (1.15)].

(iv) (The strong renewal theorem as a small-m condition [4, Section 4.1, Eq. (4.1)].) For each sign, (SRT$_\sigma$) holds if and only if

$$\lim_{\delta\to 0}\limsup_{x\to\infty} I_\sigma(\delta,x)=0, \qquad I_\sigma(\delta,x)=xT_\sigma(x)\sum_{1\le m\le \delta/T_\sigma(x)} u_m^\sigma(x). \tag{3.1}$$

(v) (Doney's local bound [2, Theorem 2.4], extending [6, Theorem A].) Under (LB$_\sigma$) there is C with $u_m^\sigma(x)\le C\,mT_\sigma(x)/x$ for all $m\ge 1$ and $x\ge a_m$.

We comment on the form of these statements. In (ii) and (iv) Caravenna and Doney use the tail $\mathbb{P}(X>x)$ and a function A with $A(x)\sim 1/\mathbb{P}(X>x)$ for renewal laws. The difference between $\mathbb{P}(X>x)$ and $T(x)=\mathbb{P}(X\ge x)$ is $f(x)=o(T(x))$. In (iv) their condition reads $\lim_{\delta\to 0}\limsup_x (x/A(x))\sum_{1\le m\le A(\delta x)} u_m(x)=0$. Since $A(\delta x)\sim \delta^\alpha/T(x)$ and the sum increases with its upper limit, this is equivalent to (3.1). (SPAN) makes the span 1, so their $\mathbb{P}(S_m\in x+(-1,0])$ is $u_m(x)$. In (v) Berger uses a scale a'_m that may differ from ours by a constant factor. For $c\,a_m\le x\le a_m$ the bound follows from (ii), since $m/a_m\le m/x$ there, so (v) holds for $x\ge a_m$ with our scale.

3.3 Elementary bounds

The following bounds use only (RV$_\alpha$) and (SPAN). We state them for the plus sign and they hold for the minus sign with the same proofs. We drop the sign from the notation.

Lemma 3.2 (sums of runs)**.** *(i) $\mathbb{P}(S_m\ge x)\le C\,mT(x)$ for all $m,x\ge 1$.*

(ii) $\mathbb{P}(S_m<x)\le e^{-mT(x)}$ for all $m\ge 0$ and $x\ge 1$.

(iii) $u_m(x)\le C\,a_m^{-1}e^{-mT(x+1)/3}$ for all $m\ge 2$ and $x\ge 0$.

Proof. (i) If $S_m\ge x$, then some $X_i\ge x$, or $\sum_i X_i\mathbf{1}_{\{X_i<x\}}\ge x$. By the union bound and Markov's inequality, $\mathbb{P}(S_m\ge x)\le mT(x)+m\,\mathbb{E}[X;X<x]/x$. Now $\mathbb{E}[X;X<x]=\sum_{1\le n<x}\mathbb{P}(n\le X<x)\le\sum_{n<x}T(n)\le CxT(x)$ by (R3).

(ii) $S_m<x$ forces $X_i<x$ for every $i\le m$, which has probability $(1-T(x))^m\le e^{-mT(x)}$.

(iii) Let $h=\lfloor m/2\rfloor\ge m/3$. Conditioning on S_h, $u_m(x)=\sum_{y\le x}u_h(y)u_{m-h}(x-y)\le \mathbb{P}(S_h\le x)\sup_z u_{m-h}(z)$. By (ii) $\mathbb{P}(S_h\le x)=\mathbb{P}(S_h<x+1)\le e^{-mT(x+1)/3}$, and by Lemma 3.1(i) $\sup_z u_{m-h}(z)\le C/a_{m-h}\le C/a_m$, since $m-h\ge m/2$. □

Lemma 3.3 (the covering kernel at scale N)**.** *Fix $\delta\in(0,1)$. Uniformly for $t\in[\delta,1/\delta]$ and $x\in[\delta,1]$,*

$$q^+_{m_N(t)+1}(\lfloor xN\rfloor)=T_+(N)\big[Q_t(x)+o(1)\big], \qquad q^-_{m_N(t)}(\lfloor xN\rfloor)=T_+(N)\big[Q^b_t(x)+o(1)\big].$$

Proof. Write $m=m_N(t)$ and $k=\lfloor xN\rfloor$, and fix $\theta\in(0,\delta/2)$. Split $q^+_{m+1}(k)=\sum_{r<k}u^+_m(r)T_+(k-r)$ at $r=k-\theta N$. On $r\le k-\theta N$ we have $(k-r)/N\in[\theta,1]$, so by (R1) $T_+(k-r)/T_+(N)=((k-r)/N)^{-\alpha}+o(1)$ uniformly. By Lemma 3.1(i), $u^+_m(r)=a_m^{-1}[g(r/a_m)+o(1)]$ uniformly in r. The error terms contribute at most $o(1/a_m)\cdot k\cdot T_+(\theta N)=o(T_+(N))$, since N/a_m is bounded. The main term is $T_+(N)\sum_{r\le k-\theta N}a_m^{-1}g(r/a_m)((k-r)/N)^{-\alpha}$, a Riemann sum with mesh $1/N$ in $z=r/N$. Since $a_m/N\to t^{1/\alpha}$ uniformly and g is bounded and uniformly continuous, it converges to $\int_0^{x-\theta}p_t(z)(x-z)^{-\alpha}dz$, uniformly in t and x. On $r>k-\theta N$, Lemma 3.1(i) and (R3) give at most $(C/a_m)\sum_{i<\theta N+1}T_+(i)\le C_\delta\,\theta^{(1-\alpha)/2}T_+(N)$ for large N, using (R1) and (R2) to compare $\theta NT_+(\theta N)$ with $NT_+(N)$. By the first bound in (2.2) the integral over $[x-\theta,x]$ is at most $C_\delta\theta^{1-\alpha}$. Let $N\to\infty$ and then $\theta\to 0$. The minus kernel is the same with u^-_{m-1} in place of u^+_m, the density g_b in place of g, and $T_-(N)\sim bT_+(N)$. The shift from m to $m-1$ does not change the limit. □

The next bound is the key estimate of the paper. It bounds the covering kernel by the tail at the level, uniformly in the number of runs, under regular variation alone.

Lemma 3.4 (the key bound)**.** *For both signs there is C with $q^{\sigma}_{m+1}(k) \le C\,T_\sigma(k)$ for all $m \ge 0$ and $k \ge 1$.*

Proof. We drop the sign. For $m = 0$, $q_1 = T$. Let $m \ge 1$. By (2.6), $q_{m+1}(k) = \sum_{r<k} u_m(r)T(k-r)$. The part with $r < k/2$ is at most $T(k/2) \le CT(k)$ by (R2). For $r \ge k/2$ put $i = k - r \in [1, k/2]$, and split the range of i into dyadic blocks $[w/2, w)$ with $w = 2^l$, $l \ge 1$, $w \le k$. On a block,

$$\sum_{w/2 \le i < w} u_m(k-i)T(i) \le T(w/2)\,\mathbb{P}\big(S_m \in (k-w, k-w/2]\big).$$

The window has at most w lattice points, all at least $k/2$. By Lemma 3.1(ii) and (R2) each point has probability at most $C(m/a_m)T(k)$. By Lemma 3.2(i) and (R2) the whole window has probability at most $C\,mT(k)$. So the block is at most $C\,mT(k)\,T(w/2)\min\{1, w/a_m\}$. Summing with (R4), the blocks with $w \le a_m$ give at most $C\frac{mT(k)}{a_m}\sum_{w \le a_m} wT(w/2) \le C\,mT(k)T(a_m)$, and the blocks with $w > a_m$ give at most $C\,mT(k)\sum_{w > a_m} T(w/2) \le C\,mT(k)T(a_m)$. Since $mT(a_m) \le C$, the total is at most $CT(k)$. □

Note that Lemma 3.1(ii) is used here at points $x \ge k/2$ that may lie far below a_m as well as above it, and it holds for all x.

Lemma 3.5 (decay in m)**.** *For both signs there are C, c with $q^{\sigma}_m(j) \le C\,T_\sigma(j)\,e^{-c\,mT_\sigma(j)}$ for all $m, j \ge 1$.*

Proof. We drop the sign. If $mT(j) \le 9$, this is Lemma 3.4 with $c = 1/9$. Let $mT(j) > 9$, so $m \ge 10$, and put $h = \lfloor (m-1)/2 \rfloor \ge m/3$. Splitting $S_{m-1} = S_h + (S_{m-1} - S_h)$ gives $q_m(j) = \sum_{s<j} u_h(s)\,q_{m-h}(j-s)$. For $s < j/2$ we use $q_{m-h}(j-s) \le CT(j-s) \le CT(j)$ (Lemma 3.4 and (R2)) and $\mathbb{P}(S_h < j/2) \le e^{-hT(j)} \le e^{-mT(j)/3}$ (Lemma 3.2(ii)). For $j/2 \le s < j$ we use

$$\sum_{j/2 \le s < j} u_h(s)\,q_{m-h}(j-s) \le \sup_{s<j} u_h(s) \sum_{i \le j/2} CT(i) \le C\,a_h^{-1} e^{-hT(j)/3}\,jT(j),$$

by Lemma 3.2(iii) and (R3). Since $hT(j) \ge mT(j)/3 > 3$, we have $T(j) \ge 3/h \ge T(a_h)$ for large h, so $j \le Ca_h$ and $jT(j)/a_h \le CT(j)$. Both parts are at most $CT(j)e^{-mT(j)/9}$. □

4 The local Lamperti law

4.1 The limit density

Proposition 4.1 (identification of the limit)**.** *The integral Λ of (2.3) is finite and continuous on $(0,1)$, and $\Lambda = L_{\alpha,b}$. Moreover, for every $\theta > -1$,*

$$\int_0^1 \frac{L_{\alpha,b}(v)\,dv}{1+\theta v} = \frac{(1+\theta)^{\alpha-1} + b}{(1+\theta)^{\alpha} + b}, \tag{4.1}$$

so $L_{\alpha,b}$ is a probability density with mean $1/(1+b)$.

Proof. Finiteness and continuity. Let x range over a compact subset K of $(0,1]$. Split $Q_t(x)$ at $z = x/2$. The part $z < x/2$ is at most $(x/2)^{-\alpha}$, and by (2.2) the part $z \ge x/2$ is at most $C\min\{t^{-1/\alpha}, t\}\int_0^{x/2} w^{-\alpha}dw$. So $Q_t(x) \le C_K(1 + \min\{t^{-1/\alpha}, t\})$, and the same holds for Q^b_t. By (2.2), $p_t(x) \le C_K \min\{t^{-1/\alpha}, t\}$. The integrand of (2.3) is therefore at most $C_K \min\{t^{-1/\alpha}, t\}$ for $x \in K$, which is integrable on $(0,\infty)$ because $1/\alpha > 1$. The integrand is continuous in x, so Λ is finite and continuous by dominated convergence.

Homogeneity. For $x, y > 0$ put $\lambda(x,y) = \int_0^\infty [p_{bt}(y)Q_t(x) + p_t(x)Q_t^b(y)]\,dt$, so $\Lambda(x) = \lambda(x, 1-x)$. For $c > 0$ the definitions give $p_t(cz) = c^{-1}p_{tc^{-\alpha}}(z)$, and the substitution $z = cw$ in (2.1) gives $Q_t(cx) = c^{-\alpha}Q_{tc^{-\alpha}}(x)$ and $Q_t^b(cy) = c^{-\alpha}Q^b_{tc^{-\alpha}}(y)$. Substituting $s = tc^{-\alpha}$ in the integral over t then gives $\lambda(cx, cy) = c^{-1}\lambda(x,y)$. Hence

$$\lambda(x,y) = \frac{1}{x+y}\,\Lambda\Big(\frac{x}{x+y}\Big).$$

The double Laplace transform. For $\lambda, \mu > 0$, $\int_0^\infty e^{-\mu y}p_{bt}(y)\,dy = e^{-bt\Gamma_\alpha \mu^\alpha}$, and since Q_t is the convolution of p_t with $w^{-\alpha}$ on $(0,\infty)$,

$$\int_0^\infty e^{-\lambda x}Q_t(x)\,dx = e^{-t\Gamma_\alpha\lambda^\alpha}\,\Gamma_\alpha\lambda^{\alpha-1}, \qquad \int_0^\infty e^{-\mu y}Q_t^b(y)\,dy = b\,e^{-bt\Gamma_\alpha\mu^\alpha}\,\Gamma_\alpha\mu^{\alpha-1}.$$

All functions are nonnegative, so by Tonelli's theorem

$$\int_0^\infty\!\!\int_0^\infty e^{-\lambda x-\mu y}\lambda(x,y)\,dx\,dy = \int_0^\infty \Gamma_\alpha\big(\lambda^{\alpha-1} + b\mu^{\alpha-1}\big)e^{-t\Gamma_\alpha(\lambda^\alpha+b\mu^\alpha)}\,dt = \frac{\lambda^{\alpha-1}+b\mu^{\alpha-1}}{\lambda^\alpha + b\mu^\alpha}.$$

Now change variables to $x = sv$, $y = s(1-v)$ with $s > 0$ and $v \in (0,1)$, so $dx\,dy = s\,ds\,dv$. By homogeneity the left side is

$$\int_0^1 \Lambda(v)\int_0^\infty e^{-s(\lambda v+\mu(1-v))}\,ds\,dv = \int_0^1 \frac{\Lambda(v)\,dv}{\lambda v + \mu(1-v)}.$$

With $\mu = 1$ and $\lambda = 1+\theta$ this is (4.1) with Λ in place of $L_{\alpha,b}$. At $\theta = 0$ it gives $\int_0^1 \Lambda = 1$.

Inversion. For $z \in \mathbb{C}\setminus[0,1]$ let $G(z) = \int_0^1 \Lambda(v)\,dv/(z-v)$, and let $H(w) = (w^{\alpha-1}+b)/(w^\alpha+b)$ with principal powers on $\mathbb{C}\setminus(-\infty,0]$. The denominator does not vanish there, since $|\arg w^\alpha| < \pi\alpha < \pi$. For real $z > 1$ put $\theta = -1/z \in (-1,0)$. Then $1/(z-v) = z^{-1}/(1+\theta v)$, so the transform gives $G(z) = z^{-1}H(1-1/z)$. Since $1-1/z \in (-\infty,0]$ exactly when $z \in (0,1]$, the right side is analytic on $\mathbb{C}\setminus[0,1]$, as is G. The two agree on $(1,\infty)$, hence on all of $\mathbb{C}\setminus[0,1]$. Since Λ is continuous and integrable, $-\frac{1}{\pi}\mathrm{Im}\,G(x+i\varepsilon)$ tends to $\Lambda(x)$ as $\varepsilon \downarrow 0$ for every $x \in (0,1)$ (Lemma A.2).

We compute the boundary value. Let $x \in (0,1)$ and $w = (1-x)/x$. As $\varepsilon \downarrow 0$, $1-1/(x+i\varepsilon)$ tends to $-w$ from the upper half plane, so it tends to $we^{i\pi}$ on the principal branch. With $E = e^{i\pi\alpha}$ this gives $(we^{i\pi})^\alpha = w^\alpha E$ and $(we^{i\pi})^{\alpha-1} = -w^{\alpha-1}E$, so

$$G(x+i0) = \frac{1}{x}\,\frac{b - w^{\alpha-1}E}{b + w^\alpha E} = \frac{1}{x}\,\frac{(b - w^{\alpha-1}E)(b + w^\alpha \bar{E})}{b^2 + 2bw^\alpha\cos\pi\alpha + w^{2\alpha}}.$$

The imaginary part of the numerator is $-b\sin(\pi\alpha)\,w^{\alpha-1}(1+w)$. Hence

$$-\frac{1}{\pi}\,\mathrm{Im}\,G(x+i0) = \frac{\sin\pi\alpha}{\pi}\,\frac{b\,w^{\alpha-1}(1+w)}{x\,(b^2 + 2bw^\alpha\cos\pi\alpha + w^{2\alpha})}.$$

Since $1+w = 1/x$, we have $w^{\alpha-1}(1+w)/x = x^{-1-\alpha}(1-x)^{\alpha-1}$, and the denominator is $x^{-2\alpha}[b^2x^{2\alpha} + 2b\cos(\pi\alpha)x^\alpha(1-x)^\alpha + (1-x)^{2\alpha}]$. So the right side is $L_{\alpha,b}(x)$, and $\Lambda = L_{\alpha,b}$. Finally, differentiating (4.1) at $\theta = 0$ gives $-\int_0^1 vL_{\alpha,b}(v)\,dv = -1/(1+b)$. □

The mean $1/(1+b)$ has the expected direction. If $b < 1$, the plus runs are heavier and the plus steps occupy more than half of the time on average.

4.2 Proof of Theorem A

We first describe the structure of the proof. By the fresh-plus identity, $P_N(k)$ is a sum over the number m of completed minus runs (first sum) or plus runs (second sum) of a product of a point probability and a covering probability. Typical configurations have m of order $1/T_+(N)$, since m runs of a law with tail T have total length of order a_m, and $a_m \asymp N$ exactly when $mT_+(N) \asymp 1$. On this range Gnedenko's theorem and Lemma 3.3 turn the sum into a Riemann sum for Λ. Large m is negligible because N steps rarely contain many more runs than $1/T(N)$, which is Lemma 3.2(iii). Small m is the delicate range. There a few runs have total length of order N, so one run is of order N, and the probability that a sum of few runs lands exactly on a given point is the quantity controlled by the strong renewal theorem. The key bound of Lemma 3.4 removes the covering probability from this range at the cost of a constant, and Lemma 3.1(iv) then says that what is left is negligible if and only if the strong renewal theorem holds. Since every term is nonnegative, the same comparison runs in the other direction and gives the necessity.

We prove the theorem for a fresh plus start. Fix $0 < x_1 < x_2 < 1$ and let k range over $[x_1 N, x_2 N]$. Put $j = N - k$, $x = k/N$ and $y = 1 - x$. By Proposition 2.1,

$$P_N(k) = \underbrace{\sum_{m\geq 1} u_m^-(j)\, q_{m+1}^+(k)}_{\mathrm{I}} + \underbrace{\sum_{m\geq 1} u_m^+(k)\, q_m^-(j)}_{\mathrm{II}}.$$

Every term is nonnegative. Fix $\delta \in (0,1)$ and split the sums over m into small $m < \delta/T_+(N)$, medium $\delta/T_+(N) \leq m \leq 1/(\delta T_+(N))$ and large $m > 1/(\delta T_+(N))$. We write $\mathrm{I_s}$, $\mathrm{I_m}$, $\mathrm{I_l}$ and so on. Let

$$\Lambda_\delta(x) = \int_\delta^{1/\delta} \big[p_{bt}(1-x)Q_t(x) + p_t(x)Q_t^b(1-x)\big]\, dt.$$

By Proposition 4.1, Λ_δ increases to $L_{\alpha,b}$ as $\delta \to 0$, and by Dini's theorem the convergence is uniform on $[x_1, x_2]$.

Medium m. Every medium m is $m_N(t)$ for $t = mT_+(N) \in [\delta, 1/\delta]$, and $a_m = Nt^{1/\alpha}(1 + o(1))$ uniformly. By Lemma 3.1(i), $Nu_m^-(j) = p_{bt}(y) + o(1)$ and $Nu_m^+(k) = p_t(x) + o(1)$ uniformly. By Lemma 3.3, $q_{m+1}^+(k) = T_+(N)[Q_t(x) + o(1)]$ and $q_m^-(j) = T_+(N)[Q_t^b(y) + o(1)]$ uniformly. Put $h(t,x) = p_{bt}(1-x)Q_t(x) + p_t(x)Q_t^b(1-x)$. Since p_t, Q_t and Q_t^b are bounded on the relevant compact sets, the products of the error terms are $o(1)$ uniformly, so

$$N(\mathrm{I_m} + \mathrm{II_m}) = \sum_{\delta/T_+(N) \leq m \leq 1/(\delta T_+(N))} T_+(N)\big[h(mT_+(N), x) + o(1)\big].$$

The number of terms is at most $1/(\delta T_+(N))$, so the error terms add up to $o(1)$. The function h is continuous on $[\delta, 1/\delta] \times [x_1, x_2]$. Indeed g is continuous and bounded, so $(t,z) \mapsto p_t(z)$ is continuous on $(0,\infty)^2$, and $(t,x) \mapsto Q_t(x)$ is continuous by dominated convergence, after the substitution $z = xw$ in (2.1), which gives $Q_t(x) = x^{1-\alpha} \int_0^1 p_t(xw)(1-w)^{-\alpha} dw$. So h is uniformly continuous there, and the sum is a Riemann sum with mesh $T_+(N)$ that converges to $\int_\delta^{1/\delta} h(t,x)\, dt$ uniformly in x. Hence

$$\sup_{x_1 N \leq k \leq x_2 N} \big|N(\mathrm{I_m} + \mathrm{II_m}) - \Lambda_\delta(k/N)\big| \to 0. \tag{4.2}$$

Large m. By Lemma 3.4 and (R2), $q_{m+1}^+(k) \leq CT_+(k) \leq C_{x_1} T_+(N)$. By Lemma 3.2(iii) for the minus sign, $u_m^-(j) \leq C a_m^{-1} e^{-mT_-(j+1)/3}$, and $T_-(j+1) \geq cT_+(N)$. For $m \geq 1/T_+(N)$ we have $a_m \geq cN$. Hence

$$N\,\mathrm{I_l} \leq C\,T_+(N) \sum_{m > 1/(\delta T_+(N))} e^{-cmT_+(N)} \leq C\, e^{-c/\delta}.$$

For $\mathrm{II_l}$, Lemma 3.5 gives $q^-_m(j) \le CT_+(N)e^{-cmT_+(N)}$, and $u^+_m(k) \le C/a_m \le C/N$, so $N\,\mathrm{II_l} \le Ce^{-c/\delta}$ as well.

Small m, sufficiency. By Lemma 3.4, $q^+_{m+1}(k) \le C_{x_1}T_+(N)$ and $q^-_m(j) \le CT_-(j) \le C_{x_2}T_+(N)$. So

$$N\,\mathrm{I_s} \le C\,NT_+(N) \sum_{m<\delta/T_+(N)} u^-_m(j), \qquad N\,\mathrm{II_s} \le C\,NT_+(N) \sum_{m<\delta/T_+(N)} u^+_m(k).$$

Since $j \ge (1-x_2)N$, (R2) gives $NT_+(N) \le CjT_-(j)$ and $\delta/T_+(N) \le C\delta/T_-(j)$, so $N\,\mathrm{I_s} \le C\,I_-(C\delta, j)$ in the notation of (3.1). In the same way $N\,\mathrm{II_s} \le C\,I_+(C\delta, k)$. Assume (SRT$_+$) and (SRT$_-$). Since j and k are at least $\min(x_1, 1-x_2)N$, Lemma 3.1(iv) gives

$$\lim_{\delta\to 0} \limsup_{N\to\infty} \sup_{x_1N\le k\le x_2N} N(\mathrm{I_s} + \mathrm{II_s}) = 0.$$

Together with (4.2), the bound on large m and the uniform convergence of Λ_δ, letting $N \to \infty$ and then $\delta \to 0$ gives $\sup_{x_1N\le k\le x_2N} |NP_N(k) - L_{\alpha,b}(k/N)| \to 0$. This proves (iii)⇒(ii), and (ii)⇒(i) is trivial.

Small m, necessity. Assume (i), let $k_N = \lfloor x_0N \rfloor$ and $j_N = N - k_N$, and take $[x_1, x_2] \ni x_0$. For $m < \delta/T_+(N)$ the part $r < k/2$ of (2.6) gives $q^+_{m+1}(k) \ge \mathbb{P}(S^+_m < k/2)\,T_+(k)$, and by Lemma 3.2(i) $\mathbb{P}(S^+_m \ge k/2) \le CmT_+(k/2) \le C_{x_1}\delta$. So $q^+_{m+1}(k) \ge \frac12 T_+(N)$ for $\delta \le \delta_0$. Since every term is nonnegative,

$$\tfrac12 NT_+(N) \sum_{m<\delta/T_+(N)} u^-_m(j_N) \le N\,\mathrm{I_s} \le NP_N(k_N) - N(\mathrm{I_m} + \mathrm{II_m}).$$

By (i) and (4.2) the right side tends to $\varepsilon(\delta) = L_{\alpha,b}(x_0) - \Lambda_\delta(x_0)$, and $\varepsilon(\delta) \to 0$ as $\delta \to 0$. The map $N \mapsto j_N$ is nondecreasing with increments 0 or 1, so it takes every large integer value j. For $j = j_N$, (R2) gives $jT_-(j) \le CNT_+(N)$ and $\delta/(CT_-(j)) \le \delta/T_+(N)$. Hence $\limsup_{j\to\infty} I_-(\delta/C, j) \le C\varepsilon(\delta)$, and (SRT$_-$) follows from Lemma 3.1(iv). The same argument with $\mathrm{II_s}$, the bound $q^-_m(j) \ge \mathbb{P}(S^-_{m-1} < j/2)T_-(j) \ge \frac12 T_-(j)$ for small m, and the sequence k_N, which also takes every large value, gives (SRT$_+$). This proves (i)⇒(iii).

For a fresh minus start the identity of Proposition 2.1 has q^-_{m+1} and q^+_m in place of q^-_m and q^+_{m+1}. This index shift changes none of the limits and none of the bounds, and the proof is the same. □

Corollary 4.2. *Assume (RV$_\alpha$) and (SPAN) and a fresh start of either sign. If $\alpha > 1/2$, the local Lamperti law (ii) of Theorem A holds. If $\alpha \le 1/2$, it holds if and only if the condition of Lemma 3.1(iii) holds for both run laws, and (LB$_\pm$) suffices.*

Proof. Combine Theorem A with Lemma 3.1(iii). □

Without the strong renewal theorem, half of the local law survives. The failure can only show up as an excess.

Corollary 4.3 (the lower half of the local law)**.** *Assume (RV$_\alpha$) and (SPAN), and let the start be fresh of either sign. Then for all $0 < x_1 < x_2 < 1$,*

$$\liminf_{N\to\infty} \inf_{x_1N\le k\le x_2N} \big(NP_N(k) - L_{\alpha,b}(k/N)\big) \ge 0.$$

If (SRT$_+$) or (SRT$_-$) fails, then $\limsup_N NP_N(\lfloor xN \rfloor) > L_{\alpha,b}(x)$ for every $x \in (0,1)$.

Proof. The medium and large ranges of the proof of Theorem A use neither (SRT$_+$) nor (SRT$_-$). Every term of the fresh-plus identity is nonnegative, so $NP_N(k) \ge N(\mathrm{I_m} + \mathrm{II_m}) = \Lambda_\delta(k/N) + o(1)$ uniformly by (4.2), and $\Lambda_\delta \to L_{\alpha,b}$ uniformly on $[x_1, x_2]$ as $\delta \to 0$. This gives the first claim. If $\limsup_N NP_N(\lfloor xN \rfloor) \le L_{\alpha,b}(x)$ for some x, the first claim gives convergence at x, and Theorem A gives (SRT$_\pm$). □

Theorem A transfers at once to mixed fresh starts, such as the fair start, since P_N is then a convex combination of F_N^+ and F_N^-. It also gives the local law of the walk itself. The position after N steps is $W_N = 2A_N - N$. So for a fresh or mixed fresh start under (RV_α), (SPAN) and ($\mathrm{SRT}_\pm$),

$$N\,\mathbb{P}(W_N = v) = L_{\alpha,b}\Big(\frac{1+v/N}{2}\Big) + o(1)$$

uniformly over $v \equiv N \pmod 2$ with v/N in each compact subset of $(-1,1)$. The limit density of W_N/N on $(-1,1)$ is $\frac{1}{2}L_{\alpha,b}((1+\xi)/2)$, and the factor $\frac{1}{2}$ is compensated by the spacing 2 of the lattice of W_N. For $b = 1$ the density $\frac{1}{2}L_{\alpha,1}((1+\xi)/2)$ is the scaling function of the ballistic Lévy walk in [20, Eq. (61)], as one checks by putting $|\xi+1| = 2x$ and $|\xi-1| = 2(1-x)$ there.

Remark 4.4 (the global law)**.** Under ($\mathrm{SRT}_\pm$), Theorem A gives $\mathbb{P}(sN \le A_N \le s'N) \to \int_s^{s'} L_{\alpha,b}$ for $0 < s < s' < 1$ by Riemann sums, and since $L_{\alpha,b}$ has total mass 1 this gives the convergence of A_N/N in law. Proposition 4.1 identifies the limit without any input from [13]. This is a derivation of the global law only under ($\mathrm{SRT}_\pm$). The convergence in law is the classical statement [13], and we do not use it. The same argument shows that $\mathbb{P}(A_N \le \varepsilon_N N) + \mathbb{P}(A_N \ge N - \varepsilon_N N) \to 0$ for every $\varepsilon_N \to 0$ under ($\mathrm{SRT}_\pm$), since the mass of P_N on $[\delta N, (1-\delta)N]$ tends to $\int_\delta^{1-\delta} L_{\alpha,b}$. So the end layers of Section 6 dominate pointwise but carry no mass in the limit.

Remark 4.5 (where the bivariate structure went)**.** The count A_N is a functional of the bivariate renewal process (S_m^+, S_m^-), and one might expect the local law to need a bivariate strong renewal theorem. The proof shows that it does not. Lemma 3.4 bounds the covering kernel by $CT(k)$ uniformly in m, which decouples the two signs for small m. The small-m part of each sign is then the univariate quantity of Lemma 3.1(iv), and a single interior point controls both.

5 The shape of the Lamperti density

We prove Theorem B. Let α_c be the unique root of $\cos(\pi\alpha/2) = \alpha$ in $(0,1)$. The function $\alpha \mapsto \cos(\pi\alpha/2) - \alpha$ decreases strictly from 1 to -1 on $[0,1]$, so α_c exists, and $\cos(\pi\alpha/2) \ge \alpha$ exactly when $\alpha \le \alpha_c$. Numerically $\alpha_c = 0.594611644056\ldots$.

Proof of Theorem B. Let $b = 1$ and $c = \cos\pi\alpha$. By (2.4) with $\beta = 0$, $L_{\alpha,1}(x) = \frac{\sin\pi\alpha}{\pi}\,\phi(t)$ with

$$\phi(t) = \frac{1+\cosh t}{\cosh\alpha t + c}, \qquad t = \log\frac{1-x}{x}.$$

The map $x \mapsto t$ is a decreasing bijection from $(0,1)$ onto $\mathbb{R}$, so the critical points of $L_{\alpha,1}$ correspond to those of ϕ, with the same type. The function ϕ is even, so $t = 0$, that is $x = 1/2$, is always a critical point. Differentiating,

$$\phi'(t) = \frac{(1+\cosh t)\big[\tanh(t/2)(\cosh\alpha t + c) - \alpha\sinh\alpha t\big]}{(\cosh\alpha t + c)^2},$$

since $\sinh t = (1+\cosh t)\tanh(t/2)$. For $t > 0$ the sign of $\phi'(t)$ is therefore the sign of

$$\chi(t) = \cosh\alpha t + c - \alpha\sinh(\alpha t)\coth(t/2) = G(t) + c, \qquad G(t) = \cosh\alpha t - \alpha\sinh(\alpha t)\coth(t/2).$$

We claim that G is strictly increasing on $(0,\infty)$. Put $s = t/2$. Using $1 + 1/(2\sinh^2 s) = \cosh t/(2\sinh^2 s)$ and $\coth s = \sinh t/(2\sinh^2 s)$ we get

$$G'(t) = \alpha\sinh\alpha t - \alpha^2\cosh(\alpha t)\coth s + \frac{\alpha\sinh\alpha t}{2\sinh^2 s} = \frac{\alpha\, n(t)}{2\sinh^2(t/2)},$$

with $n(t) = \sinh(\alpha t)\cosh t - \alpha\cosh(\alpha t)\sinh t$. Now $n(0) = 0$ and

$$n'(t) = (1-\alpha^2)\sinh(\alpha t)\sinh t > 0 \qquad (t > 0),$$

so $n > 0$ and $G' > 0$ on $(0,\infty)$. This proves the claim. As $t \downarrow 0$, $\alpha\sinh(\alpha t)\coth(t/2) \to 2\alpha^2$, so $G(0+) = 1 - 2\alpha^2$. As $t \to \infty$, $G(t) = \frac{1-\alpha}{2}e^{\alpha t}(1+o(1)) \to \infty$. Hence χ increases strictly from

$$\chi(0+) = 1 + \cos\pi\alpha - 2\alpha^2 = 2\big(\cos(\pi\alpha/2) - \alpha\big)\big(\cos(\pi\alpha/2) + \alpha\big)$$

to $+\infty$. The second factor is positive.

If $\cos(\pi\alpha/2) \geq \alpha$, then $\chi > 0$ on $(0,\infty)$. So ϕ increases strictly on $[0,\infty)$ and, being even, decreases strictly on $(-\infty, 0]$. Its only critical point is the minimum at 0, and $L_{\alpha,1}$ decreases strictly on $(0, 1/2]$ and increases strictly on $[1/2, 1)$.

If $\cos(\pi\alpha/2) < \alpha$, then χ has exactly one zero $t^* > 0$, with $\chi < 0$ on $(0, t^*)$ and $\chi > 0$ on (t^*, ∞). So ϕ has a strict local maximum at 0 and strict minima at $\pm t^*$, and no other critical point. In the variable x these are the maximum at $1/2$ and the minima at $x = 1/(1 + e^{\pm t^*})$. □

Near the threshold the two minima split off from $1/2$ like a pitchfork.

Proposition 5.1 (the minima near the threshold)**.** *For $\alpha > \alpha_c$ let $t^*_\alpha > 0$ be the zero of χ in the proof of Theorem B, so the minima of $L_{\alpha,1}$ are at $1/(1+e^{\pm t^*_\alpha})$. Then*

$$t^*_\alpha \sim \Big(\frac{3\,(2\alpha^2 - 1 - \cos\pi\alpha)}{\alpha^2(1-\alpha^2)}\Big)^{1/2} \sim \Big(\frac{12\,(1 + \frac{\pi}{2}\sin\frac{\pi\alpha_c}{2})}{\alpha_c(1-\alpha_c^2)}\,(\alpha - \alpha_c)\Big)^{1/2} \qquad (\alpha \downarrow \alpha_c).$$

In particular the minima are at distance $t^*_\alpha/4 + O((t^*_\alpha)^3)$ from $1/2$.

Proof. Expanding at $t = 0$, $n(t) = \frac{1}{3}\alpha(1-\alpha^2)t^3 + O(t^5)$, so $G'(t) = \frac{2}{3}\alpha^2(1-\alpha^2)\,t + O(t^3)$ and

$$\chi(t) = \chi(0+) + \kappa_\alpha t^2\big(1 + r(t,\alpha)\big), \qquad \kappa_\alpha = \tfrac{1}{3}\alpha^2(1-\alpha^2),$$

with $|r(t,\alpha)| \leq Ct^2$ for $|t| \leq 1$ and α in a neighbourhood of α_c, since χ is an analytic function of (t,α) near $(0,\alpha_c)$ and even in t. As $\alpha \downarrow \alpha_c$ we have $\chi(0+) \uparrow 0$. The difference $\chi(1) - \chi(0+) = G(1) - G(0+)$ is positive and continuous in α, so $\chi(1) > 0$ for α close to α_c, and the zero t^*_α lies in $(0,1)$. The same argument shows that $t^*_\alpha \to 0$. For each fixed $\varepsilon \in (0,1)$ the difference $\chi(\varepsilon) - \chi(0+) = G(\varepsilon) - G(0+)$ is positive and continuous in α, so $\chi(\varepsilon) > 0$ and $t^*_\alpha < \varepsilon$ for α close to α_c. Hence $r(t^*_\alpha, \alpha) = O((t^*_\alpha)^2) \to 0$. The equation $\chi(t^*_\alpha) = 0$ gives $(t^*_\alpha)^2(1 + O((t^*_\alpha)^2)) = -\chi(0+)/\kappa_\alpha$, which is the first equivalence. For the second, $-\chi(0+) = 2(\alpha - \cos\frac{\pi\alpha}{2})(\alpha + \cos\frac{\pi\alpha}{2})$, and as $\alpha \downarrow \alpha_c$ the first factor is $(1 + \frac{\pi}{2}\sin\frac{\pi\alpha_c}{2})(\alpha - \alpha_c)(1 + o(1))$ and the second tends to $2\alpha_c$. Finally $1/(1+e^t) = \frac{1}{2} - \frac{t}{4} + O(t^3)$. □

The first expression in Proposition 5.1 is within 0.01% of t^*_α at $\alpha = 0.5947$, where $t^*_\alpha = 0.0790$, and within 1.5% at $\alpha = 0.61$, where $t^*_\alpha = 1.018$ and the minima sit at $x = 0.2655$ and 0.7345.

At $\alpha = 1/2$ the theorem gives the arcsine density $L_{1/2,1}(x) = 1/(\pi\sqrt{x(1-x)})$, with its minimum at $1/2$. As $\alpha \to 1$ the two minima move to the ends, since $\chi(t) \to \cosh t - 1 - \sinh(t)\coth(t/2) = -2$ for every fixed t while $\chi(t) \to \infty$ as $t \to \infty$ for each $\alpha < 1$.

Margolin and Barkai [14, Section III] state that the transition between the U shape and the W shape of $L_{\alpha,1}$ happens at $\alpha = 0.5946\ldots$, in agreement with Theorem B. In the arXiv version of their paper we found the value but not its derivation. Zaburdaev, Denisov and Klafter [20, Section II.A.3] note that the propagator of the ballistic Lévy walk is U-shaped and becomes W-like for $\alpha \gtrsim 0.6$.

Remark 5.2 (unequal tails)**.** For $b \neq 1$ the same computation with (2.4) shows that the sign of the derivative in t is the sign of $\tanh(t/2)(\cosh(\alpha t - \beta) + c) - \alpha \sinh(\alpha t - \beta)$, which tends to $-\infty$ as $t \to -\infty$ and to $+\infty$ as $t \to \infty$. So the number of critical points of $L_{\alpha,b}$ is odd when they are all simple. Numerically it is 1 or 3. For instance at $\alpha = 0.7$ it is 3 for $b = 1$ and $b = 2$ and 1 for $b = 3$. We do not have a proof for $b \neq 1$.

6 The end layers and the global mode

Near the plus end, $A_N = N - j$ with j small means that the run in progress at time N is a plus run of length of order N. Its probability is of order $T_+(N)$, and the only question is where it started. It starts right after the j-th minus step exactly when a minus run ends there, which gives the renewal mass function $u_-(j)$. The next theorem makes this precise at fixed j under regular variation alone, and uniformly up to distance $(1 - \delta)N$ from the end under Doney's condition.

Theorem C (the end layers)**.** *Assume (RV$_\alpha$) and (SPAN).*

(a) For each fixed $j \geq 0$, a fresh plus start has $P_N(N - j) = T_+(N)u_-(j) + o(T_+(N))$, and a fresh minus start has $P_N(N - j) = T_+(N)u_-(j)\mathbf{1}_{\{j\geq 1\}} + o(T_+(N))$.

(b) Assume moreover (LB$_+$) and (SRT$_-$). Then for a fresh plus start and every $\delta \in (0, 1)$, uniformly in $0 \leq j \leq (1 - \delta)N$,

$$P_N(N - j) = T_+(N)\big[u_-(j)\,\Phi_b(j/N) + o(\omega_-(j))\big],$$

with Φ_b as in (2.5)*. For a fresh minus start the same holds with $u_-(j)\mathbf{1}_{\{j\geq 1\}}$ in place of $u_-(j)$.*

(c) Assume (LB$_\pm$) and $T_\pm(n) \sim c_\pm n^{-\alpha}$ with $c_\pm > 0$. Then for a fresh start of either sign and every sequence $J_N \to \infty$ with $J_N \leq N/2$,

$$P_N(k) = N^{-1}L_{\alpha,b}(k/N)\,(1 + o(1)) \qquad \text{uniformly over } J_N \leq k \leq N - J_N.$$

The same statements hold at the minus end with the signs exchanged.

Part (a) has a simple reading. The event $A_N = N - j$ with j fixed requires a plus run of length of order N. On this event the first N steps contain at most $2j + 2$ runs, since the j minus steps lie in at most $j + 1$ minus runs, and the proof shows that the long run is, with probability close to 1, the run in progress at time N. The tail of that run gives the factor $T_+(N)$. Before it, the minus steps that have been taken are exactly j, and they are the union of complete minus runs. So j must be a renewal epoch of the minus runs, which has probability $u_-(j)$. For a minus start the epoch $j = 0$ is excluded, since the walk begins with a minus run. In the continuous-time model with one law and a fair start, Wang, Schulz, Deng and Barkai [18, Section VII.A] found an atom with a boundary layer for the occupation time, and (a) is a lattice version of this layer for two laws at a fixed time.

In (b) the factor Φ_b is what makes the layer and the bulk agree. Under (SRT$_-$), uniformly for $\varepsilon N \leq j \leq (1 - \delta)N$,

$$T_+(N)\,u_-(j)\,\Phi_b(j/N) = N^{-1}L_{\alpha,b}(1 - j/N)\,(1 + o(1)). \tag{6.1}$$

Indeed, with $y = j/N$, (SRT$_-$) and (R1) give $T_+(N)u_-(j) = \frac{\sin \pi\alpha}{\pi b}\, y^{\alpha-1}N^{-1}(1 + o(1))$, and the identity after (2.5) finishes the check. For j bounded, $\Phi_b(j/N) \to 1$ and (b) reduces to (a). Without pure power tails the ratio $T_-(j)/T_-(N)$ keeps a slowly varying factor in the range $1 \ll j \ll N$, and (b) is the correct form.

6.1 Proof of Theorem C(a)

Let the start be fresh plus and fix j. We use Proposition 2.3. Fix $r \le j$. We show that $\mathbb{E}[T_+(N-j-S_r^+); S_r^+ \le N-j-1] = T_+(N)(1+o(1))$. On $\{S_r^+ \le N/2\}$ the ratio $T_+(N-j-S_r^+)/T_+(N)$ is at most $T_+(N/2-j)/T_+(N) \le C$ and tends to 1 almost surely, so this part is $T_+(N)(1+o(1))$ by dominated convergence. On $\{S_r^+ > N/2\}$ put $i = N-j-S_r^+$ and fix $I \ge 1$. For fixed r and $n \ge 1$, $u_r^+(n) \le r \sup_{s \ge n/r} f_+(s)$, because one of the r summands is at least n/r. Since $f_+(s) = T_+(s) - T_+(s+1) = o(T_+(s))$ by (R1), the part $i \le I$ is at most $I \max_{n \ge N/2} u_r^+(n) = o(T_+(N))$. The part $i > I$ is at most $T_+(I)\mathbb{P}(S_r^+ > N/2) \le C_r T_+(I) T_+(N)$ by Lemma 3.2(i) and (R2). Let $N \to \infty$ and then $I \to \infty$. The second sum in (2.8) has finitely many terms, each at most $C_j \max_{r \le j} u_{r+1}^+(N-j) = o(T_+(N))$ by the same bound on u_{r+1}^+. Hence $P_N(N-j) = T_+(N) \sum_{r \le j} u_r^-(j) + o(T_+(N)) = T_+(N) u_-(j) + o(T_+(N))$.

For a fresh minus start we use Proposition 2.2 with $\nu_- = f_-$. Since $F_{N-d}^+(N-j) = 0$ for $d > j$, it gives

$$P_N(N-j) = \sum_{1 \le d \le j} f_-(d)\, F_{N-d}^+(N-j).$$

By the fresh plus case, $F_{N-d}^+(N-j) = T_+(N) u_-(j-d) + o(T_+(N))$ for each $d \le j$, and $\sum_{1 \le d \le j} f_-(d) u_-(j-d) = u_-(j)$ for $j \ge 1$ by the renewal equation. For $j = 0$ the sum is empty. □

Only (RV) of T_+ is used in (a). The minus law enters only through u_-.

6.2 Proof of Theorem C(b)

Let the start be fresh plus. By (iii) of Lemma 3.1, (LB_+) implies (SRT_+), so Theorem A applies. For $\varepsilon N \le j \le (1-\delta)N$ the claim follows from Theorem A and (6.1), since $u_-(j) \asymp \omega_-(j)$ there. For $j \le j_0$ with j_0 fixed it is (a), since $\omega_-(j) \ge c_{j_0}$ and $|\Phi_b(j/N) - 1| \to 0$. Since $\Phi_b(y) = 1 + O(y^\alpha)$ and $u_-(j) \ge c\,\omega_-(j)$ for $j \ge j_0$ under (SRT_-), it remains to show the following. There is a function ζ with $\zeta(\eta) \to 0$ as $\eta \to 0$ such that for every small $\eta > 0$ there are $\varepsilon > 0$ and j_0 with

$$\limsup_{N \to \infty} \sup_{j_0 \le j \le \varepsilon N} \Big| \frac{P_N(N-j)}{T_+(N) u_-(j)} - 1 \Big| \le \zeta(\eta), \tag{6.2}$$

Let $k = N - j$ and write $P_N(k) = \mathrm{I} + \mathrm{II}$ as in Section 4.2. Put $M = \eta / T_+(N)$ and $\theta = \eta^{1/(2+2\alpha)}$. For small η we have $\eta^{1/\alpha} \le \theta/4$, since $1/\alpha > 1 > 1/(2+2\alpha)$, so $a_M \le \theta N$ for large N.

Term I, $m \le M$. We show that $q_{m+1}^+(k) = T_+(N)(1 + O(\eta^{1/2} + \theta + \varepsilon))$ uniformly in $m \le M$ and $j \le \varepsilon N$. For the lower bound, the part $r < \theta N$ of (2.6) gives $q_{m+1}^+(k) \ge \mathbb{P}(S_m^+ < \theta N) T_+(k)$, and $\mathbb{P}(S_m^+ \ge \theta N) \le C m T_+(\theta N) \le C\eta\theta^{-\alpha} \le C\eta^{1/2}$ by Lemma 3.2(i) and (R1). For the upper bound, the part $r < \theta N$ is at most $T_+(k - \theta N) \le (1+o(1))(1 - \varepsilon - \theta)^{-\alpha} T_+(N)$. For $r \ge \theta N$ we have $r \ge a_M \ge a_m$, so Lemma 3.1(v) gives $u_m^+(r) \le C m T_+(r)/r \le C\eta\theta^{-1-\alpha}/N$, and with (R3) this part is at most $C\eta\theta^{-1-\alpha} N^{-1} \sum_{i \le N} T_+(i) \le C\eta\theta^{-1-\alpha} T_+(N) = C\eta^{1/2} T_+(N)$.

Next, the tail of the renewal sum. For $m \ge 1/T_-(j)$ we have $a_m \ge cj$, so by Lemma 3.2(iii) for the minus sign and $T_-(j+1) \ge c T_-(j)$,

$$\sum_{m > M} u_m^-(j) \le \frac{C}{j} \sum_{m > M} e^{-cmT_-(j)} \le \frac{C}{jT_-(j)}\, e^{-cMT_-(j)}$$

whenever $MT_-(j) \ge 1$. Now $MT_-(j) \ge \eta T_-(\varepsilon N)/T_+(N) \to \eta b \varepsilon^{-\alpha}$. We choose $\varepsilon = \varepsilon(\eta) \le \eta$ so small that $e^{-c\eta b \varepsilon^{-\alpha}/2} \le \eta$. Since $u_-(j) \ge c/(jT_-(j))$ for $j \ge j_0$, this gives $\sum_{m > M} u_m^-(j) \le C\eta\, u_-(j)$. Hence $\sum_{1 \le m \le M} u_m^-(j) q_{m+1}^+(k) = T_+(N) u_-(j)(1 + O(\eta^{1/2}))$.

Term I, $m > M$. By Lemma 3.4, $q^+_{m+1}(k) \le CT_+(k) \le CT_+(N)$, so this part is at most $CT_+(N)\sum_{m>M} u^-_m(j) \le C\eta\, T_+(N)u_-(j)$.

Term II. By Lemma 3.5, $q^-_m(j) \le CT_-(j)e^{-cmT_-(j)}$. Choose $c' > 0$ with $a_{c'/T_+(N)} \le N/2$ for large N. For $m \le c'/T_+(N)$ we have $k \ge N/2 \ge a_m$, and Lemma 3.1(v) gives $u^+_m(k) \le CmT_+(N)/N$. This part is at most

$$C\,T_-(j)\frac{T_+(N)}{N}\sum_{m\ge 1} me^{-cmT_-(j)} \le C\frac{T_+(N)}{NT_-(j)} \le C\frac{j}{N}\,T_+(N)u_-(j) \le C\varepsilon\,T_+(N)u_-(j).$$

For $m > c'/T_+(N)$ we have $u^+_m(k) \le C/a_m \le C/N$, and this part is at most $CN^{-1}e^{-c''s}$ with $s = T_-(j)/T_+(N)$, because $T_-(j)\sum_{m>c'/T_+(N)} e^{-cmT_-(j)} \le Ce^{-cc's}$. Relative to $T_+(N)u_-(j) \ge c/(Nsj/N)$ this is at most $C(j/N)\,s\,e^{-c''s} \le C\varepsilon$.

Collecting the terms gives (6.2) with $\zeta(\eta) = C(\eta^{1/2} + \theta + \varepsilon + \eta) \le C\eta^{1/(2+2\alpha)}$. For a fresh minus start, Proposition 2.1 gives $P_N(k) = \sum_{m\ge1}[u^+_m(k)q^-_{m+1}(j) + u^-_m(j)q^+_m(k)]$. The second sum is treated as I and gives $T_+(N)\sum_{m\ge1} u^-_m(j) = T_+(N)u_-(j)$ for $j \ge 1$, and the first is treated as II. □

6.3 Proof of Theorem C(c)

By (iii) of Lemma 3.1, (LB$_\pm$) implies (SRT$_\pm$). Apply (b) at the plus end with $\delta = 1/2$. For $J_N \le j \le N/2$, (SRT$_-$) gives $u_-(j) \asymp \omega_-(j)$, so the error $o(\omega_-(j))$ is $o(u_-(j))$. With pure power tails, $T_+(N)u_-(j) = \frac{\sin\pi\alpha}{\pi}\frac{c_+N^{-\alpha}}{c_-j^{1-\alpha}}(1+o(1)) = \frac{\sin\pi\alpha}{\pi b}\,N^{-1}(j/N)^{\alpha-1}(1+o(1))$ uniformly in $j \ge J_N$, with $b = c_-/c_+$. Multiplying by $\Phi_b(j/N)$ gives $N^{-1}L_{\alpha,b}(1-j/N)(1+o(1))$. The minus end is the same with the signs exchanged. □

6.4 The global mode

Corollary 6.1 (the global mode)**.** *Assume (RV$_\alpha$), (SPAN) and (LB$_\pm$), and let the start be fresh of either sign. Let $\kappa_+(j) = \lim_N P_N(N-j)/T_+(N)$ and $\kappa_-(j) = \lim_N P_N(j)/T_-(N)$, given by Theorem C(a). There is J such that for all large N every global mode of P_N lies in $\{0,\dots,J\}\cup\{N-J,\dots,N\}$. If moreover $\max_j \kappa_+(j) > b\max_j \kappa_-(j)$ and the maximum of κ_+ is attained at a single j_+, then the global mode is $N - j_+$ for all large N.*

Proof. Both profiles tend to 0 and are not identically 0, so their maxima are positive and attained. By Theorem A, $P_N \le C/N$ on $[\varepsilon N, (1-\varepsilon)N]$. By Theorem C(b), $P_N(N-j) \le CT_+(N)\omega_-(j) \le CT_+(N)\omega_-(J)$ for $J \le j \le \varepsilon N$, and the same at the minus end. Since $NT_+(N) \to \infty$ and $\omega_\pm(J) \to 0$ as $J \to \infty$, for J large and then N large every value of P_N on $[J, N-J]$ is less than half of $\min\{T_+(N)\max\kappa_+,\, T_-(N)\max\kappa_-\}$. The values at distance less than J from the ends converge after division by $T_+(N)$ or $T_-(N)$, and $T_-(N) \sim bT_+(N)$. This gives both claims. □

For a fresh plus start, $\kappa_+ = u_-$ has $\kappa_+(0) = 1$, and the mode is at an end for every $\alpha \in (0,1)$. For $1 < \alpha < 2$ the comparison between the atoms and the bulk has a golden ratio threshold [16]. Here the end always wins, because the bulk values are of order $1/N$ and the end values of order $T(N) \gg 1/N$.

Remark 6.2 (what (b) needs)**.** We use (LB$_+$) only through Lemma 3.1(v), in the range $j \le \varepsilon N$. Under (SRT$_\pm$) alone we do not know whether (b) holds. It would follow from the bound

$$\sum_{i\le\theta N} T_+(i) \sum_{m\le\eta/T_+(N)} u^-_m(j)\,u^+_m(N-j-i) \le \zeta(\theta,\eta)\,T_+(N)u_-(j)$$

for $j \le \varepsilon N$, with $\zeta(\theta,\eta) \to 0$ as $\theta,\eta \to 0$. The bound of Lemma 3.1(ii) loses a factor of order N/j here.

7 The initial run

In this section the initial run has an arbitrary law (ν_+, ν_-) that does not depend on N, with $D < \infty$ almost surely. For $0 \le k \le N$ let

$$\mathcal{D}_N^+(k) = N \sum_{k/2 \le d \le \min(k, N-1)} \nu_+(d)\, F_{N-d}^-(k-d),$$

$$\mathcal{D}_N^-(k) = N \sum_{(N-k)/2 \le d \le \min(N-k, N-1)} \nu_-(d)\, F_{N-d}^+(k).$$

The term $\mathcal{D}_N^+(k)$ is the event that the initial run is a plus run that ends after at least half of the k plus steps. The rest of the walk is then a fresh minus walk that has few plus steps, so it sits in its own end layer, where the probabilities are of order $T(N)$ and not $1/N$.

Theorem D (the initial run)**.** *Assume (RV$_\alpha$), (SPAN) and (SRT$_\pm$).*

(a) (The defect.) Uniformly over k with k/N in each compact subset of $(0,1)$,

$$NP_N(k) = L_{\alpha,b}(k/N) + \mathcal{D}_N^+(k) + \mathcal{D}_N^-(k) + o(1).$$

(b) (A necessary condition.) If $NP_N(\lfloor x_0 N \rfloor) \to L_{\alpha,b}(x_0)$ for one $x_0 \in (0,1)$, then $nT_+(n)\,\nu_\pm(n) \to 0$ as $n \to \infty$.

(c) (A sufficient condition.) Assume moreover (LB$_\pm$). If $n\nu_\pm(n) \to 0$, then $NP_N(k) = L_{\alpha,b}(k/N) + o(1)$ uniformly over k with k/N in each compact subset of $(0,1)$. This holds in particular when ν_+ and ν_- are eventually nonincreasing.

(d) (An exact criterion.) Assume moreover (LB$_\pm$) and $u_\pm(i) > 0$ for all $i \ge 0$. Then the conclusion of (c) holds if and only if, for all $0 < x_1 < x_2 < 1$,

$$\sup_{x_1 N \le k \le x_2 N} NT_+(N) \sum_{0 \le i \le k/2} \nu_+(k-i)\,\omega_+(i) \to 0 \qquad \text{and}$$

$$\sup_{x_1 N \le k \le x_2 N} NT_+(N) \sum_{0 \le i \le (N-k)/2} \nu_-(N-k-i)\,\omega_-(i) \to 0.$$

(e) (A finite initial run that changes the bulk.) Let $c > 0$, $n_i = 2^{2^i}$, and let $\nu_+(n_i) = c/(n_i T_+(n_i))$ for $i \ge i_0$, with the rest of the mass of the initial run placed anywhere. Then for every $x_0 \in (0,1)$, along the N with $\lfloor x_0 N \rfloor \in \{n_i\}$,

$$\liminf NP_N(\lfloor x_0 N \rfloor) \ge L_{\alpha,b}(x_0) + c\, b\, x_0^{\alpha-1}(1-x_0)^{-\alpha}.$$

Under the hypotheses of (d) the condition of (b) is not sufficient. An initial run with a cluster of $\lfloor \sqrt{n_i} \rfloor$ equal masses $T_+(\sqrt{n_i})/(n_i T_+(n_i))$ just below each n_i, for $i \ge i_0$, and the rest of its mass at $d = 1$, satisfies the condition of (b), while the local law fails along the N with $\lfloor x_0 N \rfloor \in \{n_i\}$.

(f) (The end layer.) Assume only (RV$_\alpha$) and (SPAN). For each fixed $j \ge 0$,

$$P_N(N-j) = Q_+(N)\mathbf{1}_{\{j=0\}} + T_+(N)\kappa_+(j) + \sum_{1 \le m < N/2} \nu_+(N-m)\,\Pi_m^-(j) + o(T_+(N)),$$

where

$$\kappa_+(j) = \pi_+ u_-(j)\mathbf{1}_{\{j \ge 1\}} + \sum_{d=1}^{j} \nu_-(d)\, u_-(j-d). \tag{7.1}$$

Hence $P_N(N-j)/T_+(N) \to \kappa_+(j)$ for every $j \geq 1$ if and only if

$$W_N^+ := \sum_{1 \leq m < N/2} \nu_+(N-m)\, T_+(m) = o(T_+(N)).$$

This holds if $n\nu_+(n) \to 0$, and it holds for every fresh start.

The same statements hold with the signs exchanged.

Part (b) is the first place where the infinite mean shows. For $1 < \alpha < 2$ the bulk needs no condition on the initial run on any macroscopic interior [16]. Here a first run of length close to k leaves a fresh walk with very few plus steps, whose probability is of order $T(N) \gg 1/N$. In (f) the window condition replaces the condition $\nu_+ = o(T_+)$ of the finite-mean case, because now $\sum_{m<N/2} T_+(m)$ grows like $NT_+(N)$. At $j = 0$ the end value is $P_N(N) = Q_+(N)$ exactly. So in (f) the profile value at $j = 0$ is $\lim_N Q_+(N)/T_+(N)$ when this limit exists, and not $\kappa_+(0) = 0$.

We compare the conditions. The necessary condition (b) asks that $\nu_\pm(n)$ be $o(1/(nT_+(n)))$, and the sufficient condition (c) asks that $\nu_\pm(n)$ be $o(1/n)$. The gap between them is the factor $T_+(n)$, which tends to 0. The exact criterion (d) says that what matters is a weighted window sum of $\nu_\pm$ just below the point, with the weight $\omega_\pm(i) \asymp 1/(iT(i))$, whose partial sums up to w grow like $1/T(w)$. A single spike of size $c/(nT(n))$ at n fails (b). A cluster of w equal masses $T(w)/(nT(n))$ just below n satisfies (b) and still fails (d), because the weights $\omega(i)$ over the cluster add up to about $1/(\alpha T(w))$. Under (LB$_\pm$) every initial run with a regularly varying tail $\mathbb{P}(D \geq n)$ of negative index and an eventually monotone mass function satisfies (c), whatever its index, so it changes neither the bulk nor, by (f), the end layers. Note that the initial run may be much heavier than the other runs. For instance $\nu_\pm(n) \propto n^{-1-\gamma}$ with $0 < \gamma < \alpha$ gives an initial run with a heavier tail than every later run, and under (LB$_\pm$) the local Lamperti law still holds. The effect of such a run is only felt at speed $\mathbb{P}(D \geq \varepsilon N)$, as the computations of Section 9 show for $\gamma = 0.2$ and $\alpha = 1/2$.

7.1 Proofs

Proof of (a). Fix $0 < x_1 < x_2 < 1$ and let $k \in [x_1 N, x_2 N]$. In Proposition 2.2 the atoms vanish. Consider the plus part $\sum_{d<N} \nu_+(d) F^-_{N-d}(k-d)$, where only $d \leq k$ contributes. Fix a small $\varepsilon > 0$. For $d < \varepsilon N$ the horizon is $n = N - d \geq (1-\varepsilon)N$ and $(k-d)/n \in [x_1/2, x_2]$. Theorem A for the fresh minus walk holds uniformly in the horizon $n \to \infty$ and in the ratio in a compact set. Since $|(k-d)/n - k/N| \leq 2\varepsilon$ and $L_{\alpha,b}$ is uniformly continuous on $[x_1/2, x_2]$, this gives $NF^-_{N-d}(k-d) = L_{\alpha,b}(k/N) + O(\varepsilon) + o(1)$. So the terms with $d < \varepsilon N$ contribute $\mathbb{P}(\Sigma = +, D < \varepsilon N) L_{\alpha,b}(k/N) + O(\varepsilon) + o(1) = \pi_+ L_{\alpha,b}(k/N) + O(\varepsilon) + o(1)$. For $\varepsilon N \leq d < k/2$ the ratio $(k-d)/(N-d)$ lies in $[x_1/2, x_2]$ and $N - d \geq N/2$, so $NF^-_{N-d}(k-d) \leq C$ by Theorem A, and these terms contribute at most $C\mathbb{P}(D \geq \varepsilon N) \to 0$. The terms with $k/2 \leq d \leq k$ form $\mathcal{D}^+_N(k)$. The minus part $\sum_{d<N} \nu_-(d) F^+_{N-d}(k)$ is the same, with $d \leq N - k$ and the ratio $k/(N-d) \leq 2x_2/(1+x_2) < 1$ for $d < (N-k)/2$. Since $\pi_+ + \pi_- = 1$, letting $N \to \infty$ and then $\varepsilon \to 0$ proves (a). □

Proof of (b). All terms of $\mathcal{D}^\pm_N$ are nonnegative. Since $F^-_{N-k}(0) = T_-(N-k)$, the term $d = k$ of $\mathcal{D}^+_N(k)$ is $N\nu_+(k)T_-(N-k)$. In the same way the term $d = N - k$ of $\mathcal{D}^-_N(k)$ is $N\nu_-(N-k)T_+(k)$. By (a) and the hypothesis both tend to 0 along $k = k_N = \lfloor x_0 N \rfloor$. Both k_N and $N - k_N$ take every large integer value, and $NT_-(N-k_N)$ and $NT_+(k_N)$ are comparable with $k_N T_+(k_N)$ and $(N-k_N)T_+(N-k_N)$ by (R2). Hence $nT_+(n)\nu_\pm(n) \to 0$. □

Proof of (c) and (d). Write $i = k - d$, so $\mathcal{D}^+_N(k) = N \sum_{0 \leq i \leq k/2} \nu_+(k-i) F^-_{N-k+i}(i)$. The horizon $n = N - k + i$ lies between $(1-x_2)N$ and N, and $i/n \leq (k/2)/(N - k/2) \leq$

$x_2/(2-x_2) < 1$. Theorem C(b) at the minus end, with the signs exchanged, applies to the fresh minus walk under (LB$_-$) and (SRT$_+$). It gives, uniformly in these i,

$$F_n^-(i) = T_-(n)\big[u_+(i)\Phi_{1/b}(i/n) + o(\omega_+(i))\big].$$

Since $u_+ \le C\omega_+$, $\Phi_{1/b}$ is bounded on $[0, x_2/(2-x_2)]$ and $T_-(n) \le CT_+(N)$, we get $\mathcal{D}_N^+(k) \le C\,NT_+(N)\sum_{0\le i\le k/2}\nu_+(k-i)\omega_+(i)$. If moreover $u_+(i) > 0$ for all i, then $u_+ \ge c\,\omega_+$ by (SRT$_+$), and $\Phi_{1/b}$ is bounded below on that interval, so the reverse inequality holds with a smaller constant for large N. The same holds for $\mathcal{D}_N^-$. Since $NP_N(k) - L_{\alpha,b}(k/N) = \mathcal{D}_N^+(k) + \mathcal{D}_N^-(k) + o(1)$ uniformly with $\mathcal{D}_N^\pm \ge 0$, this proves (d).

For (c), let $\nu_+(d) \le \epsilon/d$ for $d \ge x_1 N/2$. Then by (R3)

$$NT_+(N)\sum_{0\le i\le k/2}\nu_+(k-i)\omega_+(i) \le \frac{2\epsilon NT_+(N)}{k}\Big(1 + \sum_{1\le i\le k}\frac{1}{iT_+(i)}\Big) \le C\epsilon\,\frac{NT_+(N)}{kT_+(k)} \le C_{x_1}\epsilon.$$

The same bound holds for the minus sum, so $\mathcal{D}_N^\pm \to 0$ uniformly and (a) gives the local law. If ν_+ is eventually nonincreasing, then $\frac{n}{2}\nu_+(n) \le \sum_{n/2<d\le n}\nu_+(d) \to 0$. □

Proof of (e). The masses $c/(n_iT_+(n_i))$ are summable, since $1/(nT_+(n))$ is regularly varying with the negative index $\alpha - 1$ and n_i grows doubly exponentially. So the initial run is well defined for i_0 large. Let $k = \lfloor x_0 N\rfloor = n_i$. By (a) and the nonnegativity of the other terms, $NP_N(k) \ge L_{\alpha,b}(k/N) + N\nu_+(k)T_-(N-k) + o(1)$. Now

$$N\nu_+(k)T_-(N-k) = c\,\frac{N}{k}\,\frac{T_-(N-k)}{T_+(k)} \to c\,x_0^{-1}\,b\Big(\frac{1-x_0}{x_0}\Big)^{-\alpha} = c\,b\,x_0^{\alpha-1}(1-x_0)^{-\alpha},$$

by (R1). For the second example put $n = n_i$ and $w = \lfloor\sqrt{n}\rfloor$, and let $\nu_+(d) = T_+(w)/(nT_+(n))$ for $n - w < d \le n$. The mass of the cluster is $wT_+(w)/(nT_+(n))$, which is regularly varying in n with index $-(1-\alpha)/2 < 0$, so the clusters have summable mass. For d in the cluster, $dT_+(d)\nu_+(d) \le CT_+(w) \to 0$, so the condition of (b) holds. At $k = n$ the first sum in (d) is at least

$$NT_+(N)\,\frac{T_+(w)}{nT_+(n)}\sum_{0\le i<w}\omega_+(i) \ge c\,\frac{NT_+(N)}{nT_+(n)}\cdot T_+(w)\cdot\frac{1}{\alpha T_+(w)}(1+o(1)) \ge c' > 0$$

by (R3), so the criterion of (d) fails and the local law fails along these N. □

Proof of (f). By Proposition 2.2, with $F_{N-d}^-(N-j-d) = \Pi_{N-d}^-(j)$,

$$P_N(N-j) = Q_+(N)\mathbf{1}_{\{j=0\}} + \sum_{d<N}\nu_+(d)\,\Pi_{N-d}^-(j) + \sum_{1\le d\le j}\nu_-(d)\,F_{N-d}^+(N-j),$$

since $F_{N-d}^+(N-j) = 0$ for $d > j$. By Theorem C(a) for a fresh plus walk, the last sum is $T_+(N)\sum_{d\le j}\nu_-(d)u_-(j-d) + o(T_+(N))$. In the middle sum, split at $d = N/2$. For $d \le N/2$, Theorem C(a) for a fresh minus walk gives $\Pi_n^-(j) = T_+(n)[u_-(j)\mathbf{1}_{\{j\ge1\}} + o(1)]$ uniformly in $n \ge N/2$. Since $1 \le T_+(N-d)/T_+(N) \le C$ for $d \le N/2$ and the ratio tends to 1 for each fixed d, dominated convergence gives $\sum_{d\le N/2}\nu_+(d)T_+(N-d) = \pi_+T_+(N)(1+o(1))$. So the terms with $d \le N/2$ give $T_+(N)\pi_+u_-(j)\mathbf{1}_{\{j\ge1\}} + o(T_+(N))$. The terms with $d > N/2$ are exactly $\sum_{1\le m<N/2}\nu_+(N-m)\Pi_m^-(j)$ with $m = N-d$. This proves the formula.

For the criterion, let $j \ge 1$. By Theorem C(a) for a fresh minus walk, $\Pi_m^-(j) \le C_jT_+(m)$ for all m, so $W_N^+ = o(T_+(N))$ makes the sum $o(T_+(N))$. Conversely, suppose that $P_N(N-j)/T_+(N) \to \kappa_+(j)$ for every $j \ge 1$. Then the sum, which is nonnegative, is $o(T_+(N))$ for every $j \ge 1$. Pick $j^* \ge 1$ with $u_-(j^*) > 0$. Then $\Pi_m^-(j^*) \ge cT_+(m)$ for $m \ge m^*$, so $\sum_{m^*\le m<N/2}\nu_+(N-m)T_+(m) = o(T_+(N))$. For each $m < m^*$ take

$j = m$. Then $\Pi^-_m(m) = T_-(m) > 0$, so $\nu_+(N-m) = o(T_+(N))$. Together these give $W_N^+ = o(T_+(N))$.

If $n\nu_+(n) \to 0$, then $\nu_+(N-m) = o(1/N)$ uniformly in $m < N/2$, and $W_N^+ \le o(1/N)\sum_{m<N/2} T_+(m) = o(T_+(N))$ by (R3). For a fresh minus start $\nu_+ = 0$. For a fresh plus start, fix M. The terms $m < M$ give at most $M \max_{n \ge N-M} f_+(n) = o(T_+(N))$, and the terms $M \le m < N/2$ give at most $T_+(M)\mathbb{P}(N/2 < X_+ \le N-M) \le T_+(M)T_+(N/2) \le CT_+(M)T_+(N)$. Let $N \to \infty$ and then $M \to \infty$. □

7.2 Initial runs of length of order N

If the law of the initial run depends on N and puts mass of order 1 on lengths of order N, the local law itself changes. A first run of length zN is followed by a fresh walk of length $(1-z)N$, and Theorem A applied to that walk suggests a mixture of rescaled Lamperti densities.

Conjecture 7.1 (initial runs of length of order N)**.** *Assume (RV$_\alpha$), (SPAN) and (LB$_\pm$). Let the law $\nu^{(N)}$ of the initial run depend on N, with $N\nu_\pm^{(N)}(\lfloor zN \rfloor) \to \rho_\pm(z)$ uniformly for z in compact subsets of $(0,1)$, where $\rho_\pm$ are continuous, $\sup_N \sup_{d \ge \varepsilon N} N\nu_\pm^{(N)}(d) < \infty$ for each $\varepsilon > 0$, and $\pi^0 = \lim_{\varepsilon\to 0} \lim_N \mathbb{P}(D^{(N)} < \varepsilon N)$ exists. Then, uniformly for x in compact subsets of $(0,1)$,*

$$NP_N(\lfloor xN \rfloor) \to \pi^0 L_{\alpha,b}(x) + \int_0^x \rho_+(z)\, \frac{L_{\alpha,b}\left(\frac{x-z}{1-z}\right)}{1-z}\, dz + \int_0^{1-x} \rho_-(z)\, \frac{L_{\alpha,b}\left(\frac{x}{1-z}\right)}{1-z}\, dz.$$

The parts of the first-run decomposition with $d < \varepsilon N$ and with d at distance at least θN from k are Riemann sums that converge by Theorem A. The missing piece is the near-diagonal bound

$$\sum_{|d-k| \le \theta N} \nu_+^{(N)}(d)\, F^-_{N-d}(k-d) \le C\, \frac{\theta^\alpha}{N} \quad \text{uniformly for } k/N \text{ in a compact subset of } (0,1),$$

together with the same bound for the minus part near $d = N-k$. The remainder walk is then in its end layer, and Theorem C(b) bounds it by $CT(N)\omega(k-d)$. With $\nu^{(N)} \le C/N$ near $d = k$ this suggests the bound, but we have not written out the uniformity in the horizon and in k. For D uniform on $\{1, \dots, N/2\}$ and a fair sign, $\rho_\pm = 1$ on $(0, 1/2)$, $\pi^0 = 0$, and the right side at $x = 1/2$ is $4/\pi = 2L_{1/2,1}(1/2)$ when $\alpha = 1/2$. The computations of Section 9 agree with the conjecture in this case.

8 Sign changes of the discrete law

The density $L_{\alpha,b}$ has 1 or 3 critical points when $b = 1$. The discrete law can have more local extrema, and the extra ones sit at the ends. In this section we count the sign changes of $k \mapsto P_N(k+1) - P_N(k)$, which are the strict local extrema of P_N when no two neighbouring values are equal. For a finite or infinite real sequence (s_i) we write $V(s)$ for the number of sign changes of the sequence of its nonzero terms.

Theorem E (a lower bound)**.** *Assume (RV$_\alpha$), (SPAN) and (SRT$_\pm$), and let the initial run be such that $NP_N(k) = L_{\alpha,b}(k/N) + o(1)$ uniformly on compact subsets of $(0,1)$ (for instance a fresh start, or under (LB$_\pm$) any initial run with $n\nu_\pm(n) \to 0$) and such that $W_N^\pm = o(T_\pm(N))$. Assume that the limits*

$$\kappa_+(0) = \lim_N \frac{Q_+(N)}{T_+(N)}, \qquad \kappa_-(0) = \lim_N \frac{Q_-(N)}{T_-(N)}$$

exist and are finite, and let $\kappa_\pm(j)$ for $j \ge 1$ be the end profiles of (7.1) *and its mirror. Assume that $\kappa_\pm(j+1) \ne \kappa_\pm(j)$ for all $j \ge 0$ and that $\kappa_\pm$ is strictly decreasing from some*

point on. Let $\sigma_\pm = V\big((\kappa_\pm(j+1)-\kappa_\pm(j))_{j\ge 0}\big)$*, and assume that* $L_{\alpha,b}$ *has finitely many critical points,* σ_L *in number, each a strict local extremum. Then for all large* N*,*

$$V\big((P_N(k+1)-P_N(k))_{0\le k<N}\big) \ge \sigma_L+\sigma_++\sigma_-.$$

In the definition of $\kappa_\pm(0)$ we use the atom. By Proposition 2.2, $P_N(N)=Q_+(N)$ exactly, while the formula (7.1) gives 0 at $j=0$. If $Q_+(N)/T_+(N)$ does not converge, the value at the end point can break the pattern of the profile, and the theorem says nothing.

Proof. We use the following fact. If (s_i) is a sequence and (s_{i_l}) a subsequence, then the sequence of differences of (s_i) has at least as many sign changes as the sequence of differences of (s_{i_l}). Indeed, each nonzero difference $s_{i_{l+1}}-s_{i_l}$ is a sum of consecutive differences of (s_i), one of which has its sign, and choosing one such difference for each l gives a subsequence of the differences of (s_i) with the same sign pattern.

Let $c_1<\cdots<c_s$ be the critical points of $L_{\alpha,b}$, $s=\sigma_L$. Since $L_{\alpha,b}\to\infty$ at both ends, $L_{\alpha,b}$ decreases strictly on $(0,c_1]$ and increases strictly on $[c_s,1)$, and the critical points alternate between strict minima and strict maxima, starting and ending with a minimum. Choose $z_0\in(0,c_1)$ and $z_{s+1}\in(c_s,1)$. Then the differences of $L_{\alpha,b}(z_0), L_{\alpha,b}(c_1),\dots,L_{\alpha,b}(c_s), L_{\alpha,b}(z_{s+1})$ have the signs $-,+,-,\dots,+$, with s sign changes. Choose J such that κ_+ and κ_- are strictly decreasing on $[J-1,\infty)$. Since a strictly decreasing nonnegative sequence is positive, $\kappa_\pm(J)>0$.

Consider the subsequence of $(P_N(k))_{0\le k\le N}$ at the points

$$0,1,\dots,J,\ \lfloor z_0N\rfloor,\lfloor c_1N\rfloor,\dots,\lfloor c_sN\rfloor,\lfloor z_{s+1}N\rfloor,\ N-J,\dots,N.$$

By Theorem D(f), its mirror, and the definition of $\kappa_\pm(0)$, $P_N(N-j)/T_+(N)\to\kappa_+(j)$ and $P_N(j)/T_-(N)\to\kappa_-(j)$ for $0\le j\le J$. Since consecutive values of $\kappa_\pm$ differ, for large N the differences at the two ends have the signs of the differences of $\kappa_\pm$. These have σ_- sign changes at the minus end and σ_+ at the plus end, all at indices below J, and the differences next to $j=J$ have the sign of $\kappa_\pm(J)-\kappa_\pm(J-1)<0$. In the bulk, $NP_N(\lfloor zN\rfloor)\to L_{\alpha,b}(z)$ at each chosen point, so for large N the bulk differences have the signs $-,+,\dots,+$ with s sign changes. At the two junctions, $P_N(J)\sim T_-(N)\kappa_-(J)$ and $P_N(N-J)\sim T_+(N)\kappa_+(J)$, while the bulk values are $O(1/N)$. Since $NT_\pm(N)\to\infty$, the difference from $P_N(J)$ to $P_N(\lfloor z_0N\rfloor)$ is negative and the difference from $P_N(\lfloor z_{s+1}N\rfloor)$ to $P_N(N-J)$ is positive. Reading from left to right, the last difference at the minus end is negative, the junction is negative and the first bulk difference is negative. The last bulk difference is positive, the junction is positive, and the first difference at the plus end, which is $P_N(N-J+1)-P_N(N-J)\sim T_+(N)(\kappa_+(J-1)-\kappa_+(J))$, is positive. So the junctions add no sign change, and the subsequence has $\sigma_-+s+\sigma_+$ sign changes. The fact above finishes the proof. □

The hypothesis that $\kappa_\pm$ is eventually strictly decreasing holds in the main examples. If f_- is strictly log-convex, then by Theorem 1 of de Bruijn and Erdős [5] the renewal sequence u_- is strictly log-convex. Since $u_-(j)\to 0$ under (SRT$_-$), the ratios $u_-(j+1)/u_-(j)$, which increase, stay below 1, so u_- is strictly decreasing. The zeta law $f(n)=n^{-\alpha-1}/\zeta(\alpha+1)$ is strictly log-convex.

Example 8.1 (the fair start and the fresh start)**.** Let both run laws be the zeta law, so $b=1$, and let the start be fair. Then $Q_\pm(N)=\frac12 T(N)$, so $\kappa_\pm(0)=\frac12$, and (7.1) gives $\kappa_\pm(j)=\frac12 u(j)+\frac12\sum_{d\le j}f(d)u(j-d)=u(j)$ for $j\ge 1$. Since u is strictly decreasing, $\sigma_\pm=1$ if $u(1)=f(1)>\frac12$ and $\sigma_\pm=0$ if $f(1)<\frac12$. Now $f(1)=1/\zeta(1+\alpha)$, and $\zeta(1.7)=2.054\dots$ and $\zeta(1.9)=1.750\dots$. So Theorem E and Theorem B give at least 1 sign change for $\alpha=0.3$ and $\alpha=0.5$, at least 3 for $\alpha=0.7$, and at least $3+1+1=5$ for $\alpha=0.9$. For a fresh plus start with different laws, $\kappa_+=(1,u_-(1),\dots)$ is strictly decreasing and $\kappa_-=(0,u_+(1),u_+(2),\dots)$ has one sign change, so the bound is σ_L+0+1.

α	N	ratio at $k = N/2$	ratio at $k = N/20$	sign changes
0.3	1000, 2000, 4000	1.0852, 1.0687, 1.0555	1.1023, 1.0845, 1.0694	1, 1, 1
0.5	1000, 2000, 4000	1.0149, 1.0105, 1.0075	1.0045, 1.0036, 1.0028	1, 1, 1
0.58	1000, 2000, 4000	0.9846, 0.9863, 0.9883	1.0142, 1.0114, 1.0090	1, 1, 1
0.61	1000, 2000, 4000	0.9696, 0.9739, 0.9781	1.0282, 1.0230, 1.0185	3, 3, 3
0.7	1000, 2000, 4000	0.9023, 0.9159, 0.9285	1.1202, 1.1004, 1.0833	3, 3, 3
0.9	1000, 2000, 4000	0.4903, 0.5168, 0.5433	2.2892, 2.1277, 1.9905	5, 5, 5

Table 1: The ratio $NP_N(k)/L_{\alpha,1}(k/N)$ and the number of sign changes of the first differences of P_N, for the zeta law and the fair start.

In every case of Section 9 the computed number of sign changes equals the lower bound. We expect this in general.

Conjecture 8.2 (the exact count)**.** *Under the hypotheses of Theorem E, (LB$_\pm$), the condition $|f_\pm(n) - f_\pm(n+1)| = O(f_\pm(n)/n)$ and strictly log-convex run laws, the number of sign changes of $k \mapsto P_N(k+1) - P_N(k)$ equals $\sigma_L + \sigma_+ + \sigma_-$ for all large N.*

Two lemmas would prove it. The first is a difference form of Theorem A, $N^2[P_N(k+1) - P_N(k)] \to L'_{\alpha,b}(x)$ uniformly for $k/N \to x$ in compact subsets of $(0,1)$. The second is a difference form of Theorem C(b): for $J \le j \le \varepsilon N$ the sign of $P_N(N-j-1) - P_N(N-j)$ is that of $u_-(j+1) - u_-(j) < 0$. The second needs a difference form of the strong renewal theorem, $u(j) - u(j+1) \sim (1-\alpha)u(j)/j$, and we do not know whether it holds under these hypotheses when $\alpha \le 1/2$.

9 Computations

No proof uses a computed number. The computations check the exact identities and test the theorems at moderate N. Except where we say otherwise they use the zeta law $f(n) = n^{-\alpha-1}/\zeta(\alpha+1)$ for both signs, so $b = 1$, and the fair start. The exact law comes from (2.9). The convolution powers $u_m^{\pm}$ are computed by direct convolution, and every entry of A, B and F_N^+ is a sum of nonnegative terms, so the relative rounding error stays small. The total mass is 1 to 12 digits in every case. The identities of Section 2.5 and (2.9) agree with a sum over all sign strings for $N \le 12$, for two different laws of the two signs and a general initial run, to $2 \cdot 10^{-16}$.

Table 1 shows the ratio $NP_N(k)/L_{\alpha,1}(k/N)$ at $k = N/2$ and $k = N/20$, and the number of sign changes of $k \mapsto P_N(k+1) - P_N(k)$. At $\alpha = 1/2$ the central ratio is 1.0149, 1.0105 and 1.0075 for $N = 1000$, 2000 and 4000, and the distance to 1 shrinks by a factor close to $\sqrt{2}$ when N doubles. The convergence is slower away from $\alpha = 1/2$ and much slower as $\alpha \to 1$. At $\alpha = 0.9$ the central ratio is 0.54 at $N = 4000$. The sign changes sit at $k = N/2$ for $\alpha \le 0.58$. At $\alpha = 0.7$ and $N = 1000$ they sit at $k = 104$, 500 and 896, near the minima of $L_{0.7,1}$. At $\alpha = 0.9$ and $N = 4000$ they sit at $k = 1$, 142, 2000, 3858 and 3999. The two outer ones come from the ends, as in Example 8.1. Indeed $P_N(N)/T(N) = \frac{1}{2}$ exactly, while $P_N(N-1)/T(N)$ is 0.5754, 0.5737 and 0.5727 for $N = 1000$, 2000 and 4000, tending to $f(1) = 0.5715$. On both sides of the threshold of Theorem B the count is 1 at $\alpha = 0.58$ and 3 at $\alpha = 0.61$, for all three N. Every count in the table equals the lower bound of Theorem E.

At the end, for $\alpha = 1/2$, the ratio $P_N(N-j)/(T(N)u(j))$ is given in Table 2 for $1 \le j \le 5$. It lies in $[1.0009, 1.0040]$ at $N = 1000$ and in $[1.0004, 1.0020]$ at $N = 2000$, so the error halves when N doubles.

The ratio of $P_N(N-j)$ to $T(N)u(j)\Phi_1(j/N)$ over $1 \le j \le 0.8N$ lies in $[1.0004, 1.0148]$ at $N = 1000$ and in $[1.0002, 1.0105]$ at $N = 2000$, as Theorem C(b) predicts.

	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$
$N=1000$	1.0009	1.0018	1.0026	1.0033	1.0040
$N=2000$	1.0004	1.0009	1.0013	1.0017	1.0020

Table 2: The ratio $P_N(N-j)/(T(N)u(j))$ of Theorem C(a) for the zeta law with $\alpha = 1/2$ and the fair start.

For unequal laws we take the zeta law for the plus runs and $f_-(n) \propto (n+3)^{-\alpha-1}$ for the minus runs, so $b = \zeta(\alpha+1)/\zeta(\alpha+1,4)$, with a fresh plus start. At $\alpha = 1/2$, where $b = 2.4498$, the ratios to $L_{\alpha,b}$ at $x = 0.1, 0.3, \ldots, 0.9$ lie in $[1.0121, 1.0397]$ at $N = 1000$ and in $[1.0069, 1.0199]$ at $N = 4000$. At $\alpha = 0.7$, where $b = 3.4700$, they lie in $[0.8460, 1.2994]$ at $N = 1000$ and in $[0.8855, 1.1920]$ at $N = 4000$. In both cases the number of sign changes is 2, at $k = 1$ and at the interior minimum, as in Example 8.1.

For the Lamperti density, the root of $\cos(\pi\alpha/2) = \alpha$ is 0.594611644056836, and on a grid of $2 \cdot 10^5$ points $L_{\alpha,1}$ has 1 critical point for $\alpha \le 0.5946$ and 3 for $\alpha \ge 0.5947$, at the values we tried. The transform (4.1) agrees with direct quadrature in 30-digit arithmetic, and at $\alpha = 1/2$ the double integral (2.3) agrees with $L_{1/2,1}$ to a relative error below 10^{-13} at $x = 0.2$, 0.5 and 0.7.

For the initial run we take $\alpha = 1/2$, a fair sign and $\nu_\pm(d) = d^{-1.2}/(2\zeta(1.2))$. Then $n\nu_\pm(n) \to 0$, and Theorem D(c) gives the local law. The central ratio is 0.815, 0.836 and 0.855 for $N = 500$, 1000 and 2000. The gap to 1 decays like $N^{-0.2}$, the order of $\mathbb{P}(D \ge N)$, which enters through the first run with $\varepsilon N \le D < N$ in the proof of Theorem D(a). For D uniform on $\{1, \ldots, N/2\}$ with a fair sign, $NP_N(k)$ at $k/N = 0.3$ and 0.5 is 0.898 and 1.325 at $N = 500$ and 0.884 and 1.299 at $N = 2000$. The formula of Conjecture 7.1 gives 0.8694 and 1.2732, against the values 0.6946 and 0.6366 of $L_{1/2,1}$ for a fresh start.

The scripts and their outputs, each headed by the hash of its script, are in the folder kagey-problems/problem131/paperG of https://github.com/apemm/Kagey-Problems. The command `python -I verify_all.py` runs all checks in about a minute, and README.md lists what each script tests.

A Routine facts

This appendix proves the routine facts used in Sections 2 to 4.

A.1 Regular variation

Lemma A.1. *Let T be regularly varying with index $-\alpha \in (-1, 0)$, nonincreasing and positive.*

(i) (R4) of Section 3.1 holds.

(ii) If a is increasing with $tT(a(t)) \to 1$, then $\sup_{m \ge 1} mT(a_m) < \infty$.

(iii) With $m_N(t) = \lfloor t/T(N) \rfloor$, $a_{m_N(t)}/N \to t^{1/\alpha}$ uniformly for t in compact subsets of $(0, \infty)$.

Proof. (i) The function $\phi(w) = wT(w)$ is regularly varying with index $1 - \alpha > 0$, and it is positive, so $\inf_{w \ge 1} \phi(w) > 0$ because $\phi(w) \to \infty$. By Potter's theorem [3, Theorem 1.5.6] there is x_0 with $\phi(y)/\phi(x) \le 2(y/x)^{(1-\alpha)/2}$ for $x_0 \le y \le x$. By (R2), $2^l T(2^{l-1}) \le C\phi(2^l)$. The terms with $2^l \ge x_0$ add up to at most $C\phi(x) \sum_{l \ge 0} 2^{-l(1-\alpha)/2}$, and the finitely many terms with $2^l < x_0$ add up to a constant, which is at most $C\phi(x)$. For the second sum let $x \ge x_0$. The first term with $2^l > x$ has $2^{l-1} \in (x/2, x]$, so it is at most $T(x/2) \le CT(x)$ by (R2). The later terms have $2^{l-1} \ge x$, and Potter's bounds give $T(y)/T(x) \le 2(y/x)^{-\alpha/2}$

for $y \geq x \geq x_0$. So the sum is at most $CT(x)\sum_{l\geq 0} 2^{-l\alpha/2}$. For $1 \leq x < x_0$ the sum is at most $\sum_{l\geq 1} T(2^{l-1}) < \infty$, which is at most $CT(x_0) \leq CT(x)$.

(ii) The sequence $mT(a_m)$ converges to 1 and each term is finite.

(iii) Fix $t > 0$ and write $m = m_N(t)$. Then $mT(N) \to t$ and $mT(a_m) \to 1$, so $T(a_m)/T(N) \to 1/t$. Take a subsequence along which $a_m/N \to \lambda \in [0,\infty]$. If $\lambda \in (0,\infty)$, (R1) gives $T(a_m)/T(N) \to \lambda^{-\alpha}$, so $\lambda = t^{1/\alpha}$. If $\lambda = 0$ or $\lambda = \infty$, Potter's bounds give $T(a_m)/T(N) \to \infty$ or $\to 0$, which is impossible. So $a_m/N \to t^{1/\alpha}$. The functions $t \mapsto a_{m_N(t)}/N$ are nondecreasing and the limit is continuous and increasing, so the convergence is uniform on compact sets. □

A.2 The Poisson integral

Lemma A.2. *Let $\Lambda \geq 0$ be integrable on $(0,1)$ and continuous at $x \in (0,1)$, and let $G(z) = \int_0^1 \Lambda(v)\,dv/(z-v)$ for $z \notin [0,1]$. Then $-\frac{1}{\pi}\operatorname{Im} G(x+i\varepsilon) \to \Lambda(x)$ as $\varepsilon \downarrow 0$.*

Proof. Since $\operatorname{Im}(x+i\varepsilon-v)^{-1} = -\varepsilon/((x-v)^2+\varepsilon^2)$,

$$-\frac{1}{\pi}\operatorname{Im} G(x+i\varepsilon) = \int_0^1 \Lambda(v)\,P_\varepsilon(x-v)\,dv, \qquad P_\varepsilon(u) = \frac{\varepsilon}{\pi(u^2+\varepsilon^2)}.$$

The kernel P_ε is positive with total mass 1 on $\mathbb{R}$, and its mass outside $[-\eta,\eta]$ and its supremum there tend to 0 for each $\eta > 0$. Split the integral at $|x-v| \leq \eta$. The near part differs from $\Lambda(x)\int_{|u|\leq\eta} P_\varepsilon$ by at most $\sup_{|v-x|\leq\eta}|\Lambda(v)-\Lambda(x)|$, and the far part is at most $\sup_{|u|>\eta} P_\varepsilon(u)\int_0^1 \Lambda$. Let $\varepsilon \to 0$ and then $\eta \to 0$. □

A.3 The normalization of the minus sums

Under (RV$_\alpha$), $mT_-(a_m z) \to bz^{-\alpha}$ for every $z > 0$, by (R1) and $mT_+(a_m) \to 1$. So the sum of m minus runs, divided by a_m, converges in law to the positive stable variable whose Lévy measure has tail $bz^{-\alpha}$, which is $b^{1/\alpha}Z$. Gnedenko's local limit theorem [11, Theorem 4.2.1] holds for every norming sequence under which the sums converge, so it applies with the norming a_m and the density g_b. This is the form used in Lemma 3.1(i).

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